FACULTY OF ENGINEERING AND ARCHITECTURE

DESIGN PROJECT

GROUP 03: SERIOUS GAMES

Faculty of Engineering and Architecture

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1 INTRODUCTION

The Design Project course is part of the Master of Science in Computer Science Engineering program at Ghent University. The course is designed to immerse students in a practical, team-oriented environment that mirrors real-world software engineering projects. It emphasizes hands-on learning, allowing students to apply theoretical knowledge from previous courses while developing essential technical and soft skills, such as teamwork, communication, project management, creativity, and problem-solving. By simulating a realistic workplace environment with ambiguous guidelines, evolving teams, and iterative feedback, the course prepares students to tackle complex challenges typical in the technology industry.

In this course, students are encouraged to take the lead, learn new technical skills when needed, and learn by doing. Mistakes are seen as chances to improve, and feedback focuses on helping students get better rather than just pointing out errors. Through workshops, guidance from coaches, and organized project steps, students gain hands-on experience in designing, building, and managing a project, leading to a real, working solution by the end of the course.

Of the various proposed challenges within the course, the one central to this project is that of serious games. Serious games are a category of games whose primary purpose extends beyond entertainment. They repurpose engagement and entertainment mechanisms to achieve a “serious” objective, such as education, training, or rehabilitation, while providing an entertaining experience.

1.1 Introduction to the product

A key challenge in secondary education is connecting course material to real-world applications. This challenge is particularly pronounced in STEM subjects during the first and second years, where practical connections are less direct compared to subjects such as history. Many exercises in STEM are abstract and lack a clear context, making them harder for students to relate to.

Secondary schools are a suitable focus because students at this level can complete tasks independently while following a largely standardized curriculum across classes.

To address this challenge, Three Point Crown was developed. It is an educational platform designed to provide context for otherwise conventional or routine exercises. By embedding questions within specific narratives, students are introduced to practical applications of the concepts they learn in class, improving engagement and understanding.

1.2 Structure of the report

This report is organized according to the phases of the Design Thinking methodology. Each section builds upon the previous one, providing a coherent overview of how the Three Point Crown project evolved from an initial challenge to a fully defined concept.

The report starts with the first chapter introducing the project, its educational context, and providing an overview of the report structure.

Following this is the second chapter which documents the Design thinking phase. This phase covers several subphases, starting with Challenge Introduction and continuing through the Empathize, Define, Ideate, Prototype, and Test phases, followed by a concluding section.
The third chapter gives a detailed overview of the State-of-the-Art analysis.
In the fourth chapter, the game elements of our project are described in detail.
The fifth chapter dives deeper into the functional and non-functional requirements for the product.
ADD diagrams, white-box sequence diagrams, and the deployment diagram can be found in chapter six.
All technology choices are explained in chapter seven.
Prototype and minimal viable product are described in chapter eight.
The nineth chapter details how artificial intelligence was used throughout this project.
At the end are all used references.

Commented [SS1]: TODO: fix at end before handing in

2 DESIGN THINKING

The Design Thinking phase is the first phase of this project. Design Thinking is a human-centered, creative, and practical approach to solving problems. It focuses on understanding people's needs, working collaboratively, and turning challenges into opportunities for innovation. In this phase, students form teams, select a challenge, and generate ideas to address a problem within that challenge space. The proposed solution is then tested, evaluated, and refined based on feedback to ensure continuous improvement.

To guide us through the different phases of Design Thinking, each group was provided with a Miro board. This Miro board had sections for each of the phases to be passed through: Warm-up, Empathize, Define, Ideate, Prototype and Test. It also has a separate section dedicated to a State-Of-The-Art Analysis (SOTA), an overview of which can be found in Chapter 3 of this document. The outcomes of each of these phases are described in the sections below after an introduction to the challenge of this project.

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Commented [SS2R2]: @Saira Snauwaert

2.1 Challenge introduction

Given that the description of the Serious Game challenge did not provide a concrete problem statement or context thereof, an iterative search was conducted to define the problem statement and its context. After several rounds of defining a problem hypothesis, conducting interviews, and integrating feedback the team settled on the following problem statement and context:

Commented [FD3]: Commentaar:
"Make sure to avoid repetition compared with the introduction."

Commented [AL4]: Zo goed?

Commented [AS5]: Hier mag ook geen 'we' instaan denk ik?

> **Bridging classroom content and real-world context in STEM subjects:** The Flemish Government states in its curriculum objectives that secondary school students should be able to make connections between their learning material and the real world. Some teachers struggle with this. The issue seems to be most prominent in STEM (Science, Technology, Engineering and Mathematics) subjects in the first years of secondary school. The many (simple) exercises in these subjects often come across as quite abstract to students, because they are not strongly linked to a specific context.

Commented [AL6]: Should we try something else than a list?

Commented [SS6R2]: I think it looks fine without a number like this.

In the following sections, the different phases of the Design Thinking process are presented along with their corresponding outcomes.

2.2 Warm-up Phase

The Warm-up Phase served as an introduction to both Design Thinking and the project team. This phase helped establish a basic understanding of Design Thinking. The main objectives of this phase were to introduce the group to Design Thinking, to Miro and to each other. Furthermore, the team roles were to be decided in this phase and all members were to gain insight in the different deadlines for the project. Key takeaways from this phase included recognizing the value of each team member's unique strengths and the importance of clearly defined roles for effective collaboration.

2.3 Empathize Phase

The first actual phase of Design Thinking is the Empathize Phase. The Empathize Phase focuses on understanding the needs, frustrations, and motivations of the target audience and relevant stakeholders. This involves gathering insights through research, interviews, and observation to identify opportunities for innovation. The goal is to gain a deep understanding of user experiences and pain points before defining a specific problem.

2.3.1 Process and Outcomes of the Empathize Phase

The Empathize Phase can be split up into two main parts: a first part consisting out of defining stakeholders, interviewing them and creating personas, and a second part consisting of filling in a first version of the State-of-the-Art Analysis. The outcomes after empathizing with the stakeholders are presented below, while the State-of-the-Art Analysis is discussed in Chapter 3**Error! Reference source not found.**.

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Commented [SS7R2]: @Saira Snauwaert

The stakeholders identified for the problem statement were the following:

- Secondary school teachers of STEM subjects.
- Secondary school students from various grade levels.
- Secondary school principals.

In order to empathize with these stakeholders, questionnaires were sent out to both teachers and students. Interviews in person were also performed. The information acquired from the stakeholders, lead to the definition of the personas below.

Commented [FD8]: Commentaar:
"Be more specific. Whom did you interview to verify each persona? Are they based on multiple interviews each or only one?"

Sixteen secondary school teachers, all of whom teach STEM subjects, were surveyed. These STEM subjects included Mathematics, Physics, Natural Sciences, Technology, and ICT. Five of the interviewed teachers primarily taught the 1st year of secondary school, six of them taught the 2nd year and the others primarily taught higher years in secondary school.

All of these teachers conformed that it is important to them to link their learning material to real life contexts. Fifteen teachers said that they already pay attention to linking the real world to their subject matter in their classes, but only three teachers think that they do this frequently enough. The most important reason that teachers do not link context to exercises frequently enough is ‘difficulty to find relevant links to the real world’ (8 teachers). Another important reason is that the online course material supplied by the publisher does not spend enough time on links to the real world (4 teachers). Teachers also said that they do not have enough class-time to spend adequate time on this during their lectures (3 teachers).

When linking course material to the real world, nine teachers primarily employ exercises with a story situating it in the real world. One notable teacher primarily linked their material to forensic cases and let the students play detectives for fake murders.

Thirteen teachers considered it an added value to their curriculum if students could solve exercises with a story around it connecting it to the real world in a digital platform. They informed us from their experience that narratives connected to the exercise help students see that the subject is truly useful and make the subject matter easier to understand. When asked what exercises are most difficult to

teach and link to the real world, teachers consistently said that the easiest exercises from their subjects come across as the most vague for students.

The persona template in Figure 1 summarizes the needs, goals, tasks, actions, and design requirements emerging from the interviews and surveys.

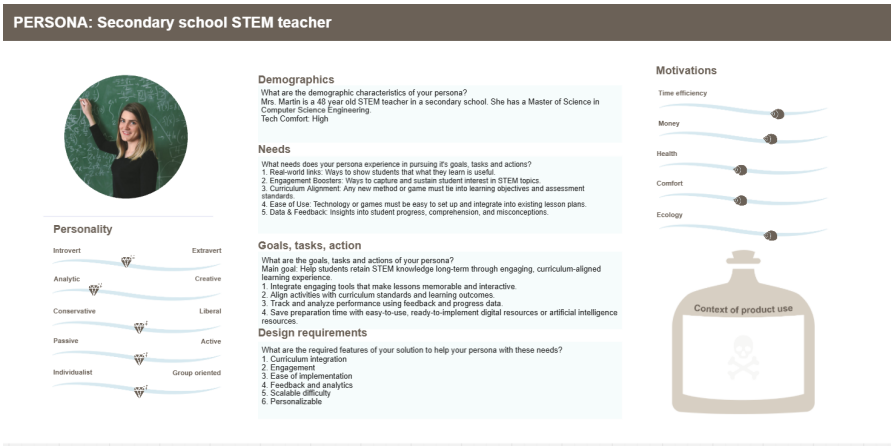


Figure 1: Persona template secondary school STEM teacher

A secondary school principal was also interviewed. The outcomes of this interview are summarized in the persona template below.

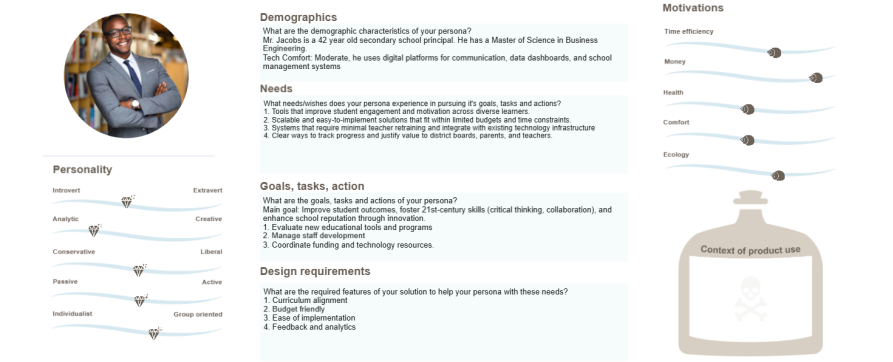


Figure 2: Persona template secondary school principal

Secondary school students were also questioned. Because of the age difference between the beginning and the end of secondary school, attention was directed to both younger and older student groups. In total, four students from the third degree and ten students from the first degree were questioned. This resulted in the two persona templates shown in Figure 3 and Figure 4.

Commented [SS9]: Er zijn 10 antwoorden op de forms, klopt dat er 10 ksa kindjes het hebben ingevuld he @Antje Schruns?

Commented [AS9R2]: Yep da klopt idd

Commented [SS9R3]: Waren dit allemaal eerste graads?

Commented [AS9R4]: Ja dat waren allemaal van 1e en 2e middelbaar

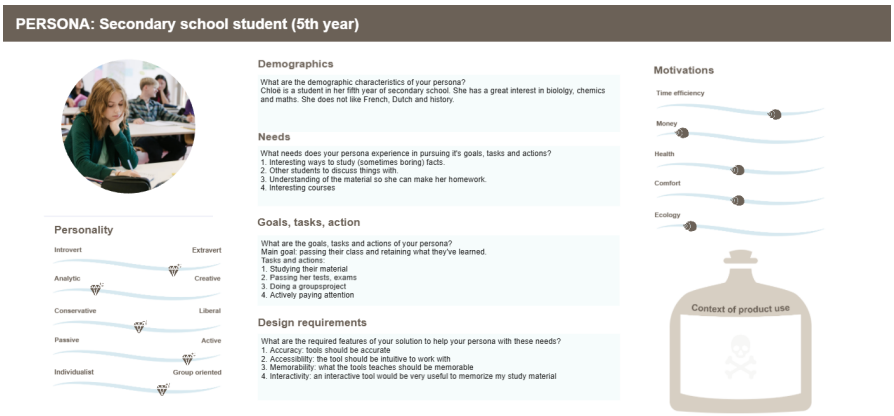


Figure 3: Persona template secondary school student (5th year)

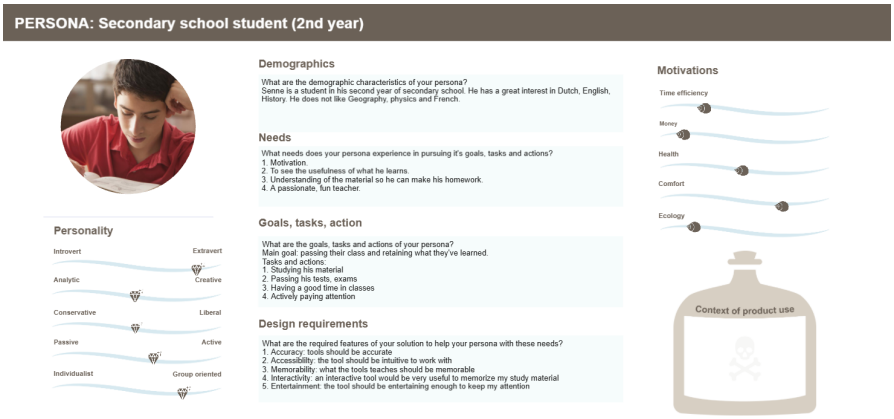


Figure 4: Persona template secondary school student (2nd year)

2.3.2 Conclusions of the Empathize Phase

Analyzing the four persona templates reveals clear areas of overlap. The key takeaways for this project were the following:

- All stakeholders value engagement and interactivity: students want learning to be fun and memorable, while teachers and the principal seek tools that sustain motivation and attention.
- Equally important is educational reliability: where students expressed the requirement for a tool to be accurate, the adult personas stressed the requirement for curriculum alignment.

- Ease of use and minimal setup were repeatedly mentioned and will thus be of great importance during the following phases of the project.
- Teachers and the principal also noted the need for feedback and analytics to monitor student progress and evaluate the tool's impact.

Addressing these shared needs can inform a solution that benefits students, teachers, and school leadership alike. Developing these persona templates based on the interviews provided valuable insight into stakeholders' needs, expectations, and design requirements, which directly guided the subsequent phases of the project.

2.4 Define Phase

In the Define Phase the research findings from the Empathize Phase are consolidated to identify the core problems faced by users. While the Empathize Phase involves exploring a broad range of user experiences and needs, this stage narrows the focus to specific and practical insights. The outcome of the Define Phase should be a clear and concise problem formulation that voices the concerns of the stakeholders.

To achieve this, the key takeaways from the persona templates of the stakeholders mentioned in the previous section were framed as "How might we" questions. The question ultimately chosen as the foundation for this project was:

"How might we help secondary school teachers link their course material to real-world applications in order to comply with their curriculum objectives and show their students the usefulness of their subject?"

This question was selected because there are not many competitors who put a similar focus on real-world applications. It is also something that we ourselves would have appreciated during our studies, and thus we have a personal interest in realizing the product.

Selecting this question as the basis for the product enables the creation of a solution for the selected problem statement. This solution will address the stakeholder needs identified in the previous section, with a focus on both teachers' requirements and students' learning experiences.

2.5 Ideate Phase

This section discusses the Ideate Phase. This next phase of Design Thinking involves developing a variety of potential solutions to meet the unmet needs of stakeholders identified in the previous phases. Having previously refined the focus to a specific problem, this phase reopens the creative process to explore multiple possible approaches to address it.

2.5.1 Process and Outcomes of the Ideate Phase

Before the brainstorming of potential solutions could get started, benchmarking was performed. This involves examining already existing solutions that address similar problems or needs. By analyzing these alternatives, it became possible to identify what works well, what does not, and where

Commented [SS10]: Is dit voldoende synthesis? Ik zie niet direct andere overlappingen

Commented [AL10R2]: Voor mij wel.

Commented [SS11]: Is dit voldoende voor de Define Phase volgens jullie? Ik weet niet goed wat nog meer te vertellen. Als ik begin over we hebben gevote blablabla dan gaan ze waarschijnlijk zeggen dat het te verhalend is.

Commented [SS11R2]: Is de outcome template nodig hier? Ik zou hem misschien wegoen omdat het eigenlijk niet veel zegt

Commented [SS11R3]: Of is het beter als ik die outcome template zet waar ik nu de vraag heb gecentreerd?

Commented [FD11R4]: Dit is idd ook een afbeelding die niet veel zegt en kan weggesmeten worden.

Commented [SS11R5]: Ja maar ik meen me te herinneren dat ik ergens las "don't forget to include your outcome templates". Ik vind alleen niet meer terug waar dat was

Commented [FD11R6]: ja heb dat ook gelezen maar mss is het gewoon dat de info die daar in staat ergens moet overgebracht worden

Commented [SS12]: Ze zeggen dat elke sectie een conclusie moet hebben, maar eigenlijk is heel deze sectie een conclusie, mag ik het hier gewoon negeren?

Commented [AS12R2]: Van mij wel, ik denk idd dat het hier weinig meerwaarde zou leveren

Commented [FD13]: Denk dat we best nog wat meer uitleg geven over waarom we voor deze vraag gekozen hebben.

Commented [FD14]: Heb even dit getypt als verklaring, feedback welkom :)

Commented [SS14R2]: Lijkt mij een correcte verklaring, maar vermoed dat er commentaar op zal komen dat dat te persoonlijk geformuleerd is. Zou het dus in de plaats passief maken

Commented [FD14R3]: Is passiever nu. Heb in de 2de zin 'we' behouden omdat dat effectief bedoeld was om persoonlijk te klinken.

Commented [SS14R4]: Dit vind ik al beter, ben echter niet zeker of het de bedoeling is dat we uitleggen waarom we secondary school kiezen aangezien dit volgt uit de initiele probleemstelling. Hetzelfde voor de real-world applications. Zou deze uitleg dus hier weglaten en dan bv meer naar voren in het document plaatsen. Nog voor we de probleemstelling verwoorden dan.

Commented [FD14R5]: zo?

Commented [SS14R6]: Jups, vind ik beter

Commented [AS15]: Vind dit een heel lange zware zin, en ik weet niet of "enables to create" correct engels is

Commented [AS15R2]: Suggestie: Selecting this question as the basis for the product enables the creation of a solution to the selected problem statement. This solution addresses the stakeholder needs identified in the previous section, with a focus on both teachers' requirements and students' learning experiences.

Commented [AL16]: Commentaar:
"There should always be an introductory paragraph separating titles, that mentions what will be discussed in section 2.5"

Commented [SS16R2]: Zo goed?

Commented [AL17]: Commentaar:
"I would discuss these in your SOTA chapter, and you can then refer to that analysis here"

improvements could be made. During these discussions, attention was focused on features that made each product stand out and could inform or inspire the development of the new design. The analysis of relevant competitors is included in the Competitive Analysis section of the SOTA chapter (Chapter 3).

During brainstorming, the COCD-box method was used to evaluate and rank different ideas according to feasibility and originality. Several of the proposed ideas were more appropriately classified as essential features for the final product. These were grouped together to serve as reference points during discussions of additional concepts. The remaining ideas were evaluated using the COCD-box axes. For example, some ideas appeared as simplified or less innovative versions of existing games, and were positioned in the top-left corner of the COCD-box.

The outcome of ideate phase is a description of the final idea and the problem it aims to solve. The team developed a clear concept that addresses the main challenges identified earlier in the Design Thinking process. The first challenge concerns the gap between the content students learn in school and its relevance in the real world. The second relates to student motivation and engagement. The proposed solution focuses on first-grade secondary school teachers and students and takes the form of a game-based platform in which teachers can upload their existing exercises. The system transforms these exercises into quests set in a virtual world.

Students control a modern prince / princess character who interacts with Non-Player Characters (NPCs) representing various professions. The reason that the player character is a prince / princess is to give them a reason for the amount of weight their decisions hold. They are not a king / queen since those roles imply a certain age and experience that is less likely to reflect the students. Students should be able to identify with their character. The NPCs pose questions based on the teacher's STEM learning content. When students answer correctly, their kingdom improves and they receive coins they can use for customization of their character and their palace. When answers are incorrect, the game world visibly deteriorates, for example through damaged buildings. In this way, the platform connects curriculum content to real-world applications while offering a more engaging learning experience.

The concept offers several benefits for its stakeholders. Teachers can more easily connect their exercises to meaningful contexts and can reuse material they already possess. Students gain insight into the real-world relevance of their study material while learning in a more motivating environment. At the same time, the ideation process highlighted several risks that require attention as development continues. The creation of a dynamic game world with many interacting elements introduces the possibility of inconsistent states, which means that state management must be designed carefully. There is also a possibility that teachers will consider the platform an additional burden rather than a helpful tool, which reinforces the need to automate processes such as exercise extraction as much as possible while maintaining accuracy.

An additional stakeholder emerged during the Ideate Phase compared to the Empathize Phase discussed in Section 2.3. During this phase, a representative from the Plantyn [1] publishing house was contacted, specifically the person responsible for STEM subjects in secondary education. The interview focused on understanding the functionalities already present in Plantyn's digital learning platform and exploring how the proposed solution could fit within existing educational practices.

Commented [AS18]: Ik zou deze zin mss wat anders formuleren want "the attention was focused" klinkt wat akward volgens mij? Ik zou "the" weglaten.

Commented [SS19]: Moeten de key takeaways hier opnieuw vermeld worden of is de verwijzing oke?

Commented [AS19R2]: Voor mij is de verwijzing oke

Commented [SS20]: TODO: update/properly reference

Commented [SS20R2]: @Saira Snauwaert

Commented [SS21]: Zou dit vervangen door iets van "Students should be able to identify with…"

Commented [FD22]: Feedback gewenst.

Commented [SS22R2]: Kleine comment, rest is goed.

A key insight from this conversation was the importance for teachers to be able to monitor student progress, an aspect strongly valued in current digital tools. The representative responded positively to the concept of linking course material to real-world context. However, caution was advised regarding the integration of AI into the platform, given that the target audience consists of students under the age of eighteen. It was also mentioned that gamification may be perceived as childish by older students, and that most use of their digital tools occurs primarily in the first grade of secondary school. This confirmed our choice to focus the solution on that age group.

Plantyn subsequently granted access to their STEM learning materials, including natural sciences, technology, and the different mathematics methods for the first. Access included e-books, teacher materials (such as test questions and supplemental exercises), and student materials (including digital assignments and knowledge clips). In return, the publishing house requested periodic updates on the project's progress, as the insights gained may also prove valuable for their own development efforts.

2.5.2 Conclusion of the Ideate Phase

The Ideate Phase resulted in a well-defined concept that addresses the core challenges identified earlier in the Design Thinking process. Through benchmarking, structured brainstorming, and idea evaluation, the team arrived at a solution that combines educational relevance with engaging gameplay. The final concept, a game-based platform that transforms teacher-provided exercises into contextualized quests, can successfully bridge the gap between abstract STEM content and real-world application while fostering student motivation.

This phase also expanded the stakeholder landscape. An interview was performed with Plantyn, whose feedback validated the chosen target audience and highlighted important considerations regarding progress tracking and the responsible use of AI in an educational context. The additional access to teaching materials further strengthened the feasibility of the proposed solution.

Overall, the Ideate Phase produced a promising and coherent direction for the subsequent design and development phases: a platform that merges learning and engagement, aligns with curricular objectives, and addresses the needs of both teachers and students.

2.6 Prototype Phase

After completing the Ideate Phase, the next step is to start prototyping. Prototypes are created in order to be able to present the idea to the stakeholders in a clear manner. Stakeholder feedback is mentioned in this section after each prototype, while the full set of gathered recommendations can be found in the test phase section.

2.6.1 Process and Outcomes of the Prototype Phase

In total, during this phase, three prototypes were created. The first two prototypes were made in parallel. These were presented to the stakeholders and using their feedback, a third prototype was created. All three are described in short below.

Commented [SS23]: Heb de indruk dat onze Prototype en Test Phase wat in elkaar overlopen. De commentaar van stakeholders hoort volgens deze intro bij Test Phase maar we hebben dat nu bij Prototype Phase. Wat doen we hiermee?

Commented [FD23R2]: Heb intro aangepast, dit ok?

Commented [FD23R3]: Heb ook de intro van de test phase aangepast om een beter beeld te geven van wat daar effectief staat.

Commented [SS23R4]: Lijkt in orde.

Commented [SS24]: Denk niet dat dit een correcte reference is

Commented [FD24R2]: Wou verwijzen naar de set of recommendations, maar heb nu plaats van reference in deze zin verplaatst om het duidelijker te maken.

Commented [FD25]: Willen we mss kort de mockups van later in het project hier vermelden als de meest recente weergave van ons idee?

Commented [SS25R2]: Ja lijkt mij nuttig.

2.6.1.1 First Prototype

The first version of the prototype was created using Figma, a collaborative design platform. This prototype had a pixel-art style and is pictured in Figure 5. During the creation of this prototype, our concept was slightly different and the player character was a king instead of a prince or princess.



Figure 5: First prototype

All of these images show the view of the student. Central in this view is the king, while the city is in the background. Different characters come up to this king with their questions. When the king answers these questions correctly, the city flourishes and he receives more money. On the other hand, if he is wrong, the city starts breaking down. The king's outfit, crown, and the throne room itself would be customizable with the coins earned from completing questions.

This prototype was presented to three students from the first grade of secondary school and two from the third grade of secondary school. The feedback they gave was:

- The font does not match the art style, pay more attention to visual coherence.
- The customization options look fun.
- The idea of the game itself seems fun and like something they would like to play.
- It is hard to see which effect is caused by each subject if the city as a whole is affected.
- If text is split up into many small text bubbles, it is difficult to remember what the full question is.
- It is good that progress is gradual and that there are many different states for the city.

Commented [SS26]: @Antje Schrurs Hoeveel en welke jaren precies hebben deze prototypes bekeken?

Commented [AS26R2]: 3 van het eerste middelbaar, 2 van het 2e middelbaar en 2 van het 5e middelbaar

Commented [AS26R3]: Maar is het nodig om dat hier te vermelden?

Commented [SS26R4]: Ja Stephanie vroeg om te specifieren wie en hoeveel er commentaar hebben gegeven.

Commented [FD27]: Was het niet ook voorgelegd aan Saira's zus die in het 5de jaar zit?

Commented [FD27R2]: En ook een vriendin van haar dacht ik? Dus nog extra 2 studenten van het 5de jaar?

2.6.1.2 Second Prototype

A second prototype was made using markers and paper and is shown in Figure 6:



Figure 6: Second prototype

In this prototype a different approach was chosen. The king character was left out entirely, so that the view of the city now became central. NPCs still come to ask the player questions, but now the customizable aspect is moved to the city. Players can choose where to put the buildings and add small decorative touches. After answering a question, the effect on the city is shown.

This prototype was also presented to the same five secondary school students. The feedback they gave was:

- It is difficult to see which effect is caused by each subject.
- The visual coherence is far better than for the first prototype.
- It is sad that there is no character customization.
- Small effects on the city can better showcase how things change.
- It is good that the city is central and not just part of the background.

Both prototypes were also presented to a teacher of mathematics and programming in the first and second year of secondary education. The main conclusions from this interview were:

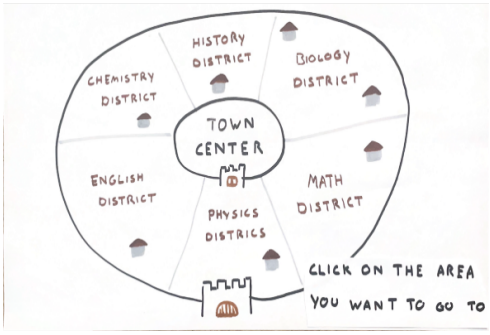
- Teachers want a clear overview of which exercises students make and how well they are doing. Being able to track student progress is of great importance.
- There should not be too much text on the screen at once, as many secondary school students will not properly read through blocks of text. It was suggested that text could appear incrementally to make it more easily digestible.
- It is important that adding exercises to the platform does not involve too much extra work from the teacher.

2.6.1.3 Third Prototype

Keeping the received feedback in mind, a third version of the prototype was created. The students perspective version was drawn by hand and is pictured below.

Commented [FD28]: @Saira Snauwaert dit ok?
Commented [SS28R2]: Zeker!

Commented [SS29]: Must be adapted to new idea + need teacher prototype
Commented [SS29R2]: @Briek Rombaut ging jij dit fiksen?
Commented [SS29R3]: Vergeet ook geen captions toe te voegen aan de afbeeldingen
Commented [BR29R4]: Sorry @Saira Snauwaert ik zie nu pas dat je me hier had getagged. Moet hier nu nog iets aan gebeuren of is dit ondertussen toch al klaar?
Commented [AS29R5]: @Briek Rombaut ik zal dat wel doen
Commented [FD29R6]: Hallo, is ons echt 3rde prototype dan volledig weg uit de deliverable?


HISTORY DISTRICT
BIOLOGY DISTRICT
CHEMISTRY DISTRICT
TOWN CENTER
ENGLISH DISTRICT
MATH DISTRICT
PHYSICS DISTRICT
CLICK ON THE AREA YOU WANT TO GO TO


PALACE
1 NEW ASSIGNMENT
LIBRARY
ROYAL SHOP
TOWN SQUARE
GUILD
MARKET
BACK


BIOLOGY DISTRICT
QUESTS
15
DOCTOR
ZOO
RESEARCH CENTER
BACK


BIOLOGY DISTRICT
QUESTS
TO DO
COMPLETED
• BLOOD GROU
• POND ECOSYSTEM
• PLANT STRUCTURE
DOCTOR
RESEARCH CENTER
BACK

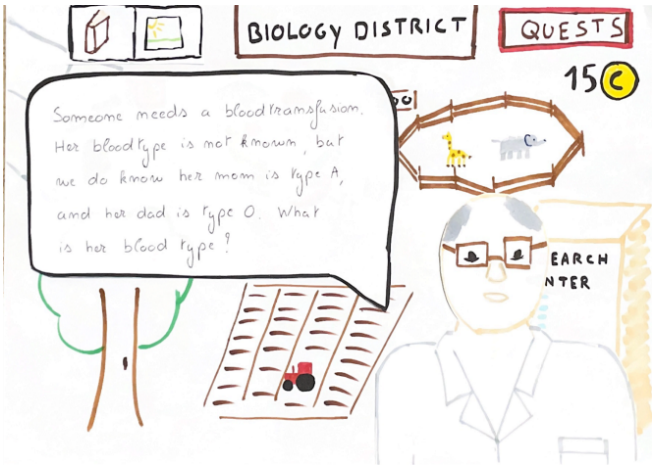

BIOLOGY DISTRICT
QUESTS
15
Someone needs a bloodtransfusion. Her bloodtype is not known, but we do know her mom is type A, and her dad is type O. What is her blood type?

Figure 7: Third prototype

For this prototype, the idea was to integrate three fundamental principles into the game: preparation, integration, and evaluation. Preparation allows students to review and organize course materials before attempting exercises. Integration gives them the opportunity to apply knowledge through interactive quests. Evaluation provides a structured way for students to demonstrate their understanding while enabling teachers to assess progress in a familiar and engaging environment.

Commented [SS30]: Leave this out?

Commented [SS30R2]: Or is it fine to leave it in and then in the conclusion below leave it out?

Commented [AS30R3]: Ik denk dat we het beter overal in of overal uit zouden laten (anders gaat er weer verwarring zijn) en voor de moment neig ik naar het overal uit laten

Commented [AS30R4]: Ofwel moeten we in de conclusie duidelijk vermelden wrm we het niet gaan doen

Commented [BR31]: Deze sectie ziet er al goed uit

Commented [AS31R2]: @Briek Rombaut vul je aan met de teacher prototype ook? En conclusie aanvullen

2.6.1.4 Fourth Prototype

After an in-depth discussion of the prototypes with both the team and coaches of the course, it was concluded that the idea presented in the third prototype incorporated too many things at once. The core of the project needed to be simplified to a more manageable scope while maintaining its focus on innovation. The refined idea is as follows:

- Teachers will be able to upload their exercises in PDF format.
- The teacher will have an easy way to convert static learning materials (PDFs) into automatically corrected exercises that align with government-defined learning goals.
- The teacher will be able to track student progress.
- A set of characters and effects will be prepared, which can be connected to real-world contexts for different STEM disciplines.
- The game will be situated in present time to more accurately reflect the world today.
- Exercises will be presented within short, contextualized scenarios or stories. For example, instead of simply solving an equation, a student might be asked to help a bank calculate whether someone has properly paid off their interest.
- The effects of the different subjects will be separated into distinct districts, so that progress in each subject can be tracked individually.
- Presenting exercises within a context makes them more engaging for students, as they are less abstract and easier to relate to.
- Since many teachers already use digital tools for homework, this approach is intended to replace existing digital tools rather than increase overall screen time.

The prototype for the refined idea can be found below: the student part was drawn by hand while the teacher side was modeled digitally.





2.6.2 Conclusions of the Prototype Phase

The prototype phase highlighted the critical importance of incorporating feedback from both students and teachers. Insights from the stakeholders helped identify usability issues, clarify visual and gameplay elements, and ensure that the platform meets educational objectives. This iterative feedback loop guided the simplification of the concept and ensured that the refined idea is both practical and aligned with the needs of its target users. For example, the separation of progress of different subjects is something that will continue to be carefully considered during further design and prototyping. From the teacher's perspective, the interface for monitoring student progress and accessing automatically corrected exercises will also be carefully optimized to be intuitive and efficient.

Commented [AS32]: Ik denk niet dat dit zo een belangrijke need is eigenlijk, kunnen we hier niet iets zetten over bvb dat de gradual progress of separation of courses? Ik denk dat het beter is om hier een relvanter voorbeeld te gebruiken

Commented [SS32R2]: Ja true, maar ergens verder zeggen we dan dat we voor mvp maar 1 district gaan uitwerken

2.7 Test Phase

Commented [FD33]: Zou hier ook de enquête dat 16 ofzo leerkrachten hebben ingevuld vermelden om onze punten van "ze zien AI wel zitten" kracht bij te stellen.

Commented [SS33R2]: Dat zit ondertussen in de Empathize phase dus zou dat niet nog eens herhalen hier.

The Test Phase focuses on gathering a set of recommendations and constructing a product vision board based on the feedback of the prototypes created during the previous stage of Design Thinking. In this phase, insights are formed by analyzing stakeholder feedback and identifying what works well with them and what needs improvement. Results from this phase help refine ideas, reveal new needs and guide further design decisions.

The ultimate goal of this phase is to come up with a set of concrete recommendations that will feed directly into the next steps of the project, i.e. the Requirements Phase. The results of this phase and how they shaped the product are summarized below.

2.7.1 Set of Recommendations

After presenting the different prototypes to the end users, in this case secondary school students and teachers, a set of recommendations was put together. The recommendations can be split into two types: functional recommendations and non-functional recommendations. Functional recommendations are high-level ideas or proposals for product improvements that focus on what the product should do. Non-functional recommendations focus on how the product performs in terms of usability, aesthetics and overall user experience.

The functional recommendations drawn from the interviews are the following:

- Provide clear visual indicators of progress for the different subjects.
- Make sure progress is gradual, that there are many steps between the best and worst state.
- Reward players with customization options and/or upgrades.
- Make the tool relevant and useful for teachers across STEM courses.

The non-functional recommendations are the following:

- Design the game to be visually appealing to encourage (initial) engagement.
- Ensure the theme is visually coherent throughout the game.
- Minimize the amount of work that teachers need to do to use their exercises on the platform.

These recommendations will be incorporated into the development of the final product to ensure it meets the needs of both students and teachers.

2.7.2 Product Vision

The vision of Three Point Crown is to make STEM education in Flemish secondary schools more tangible and enjoyable by representing real-life impact through gamification. The platform addresses the curriculum objective of linking STEM subjects with their real-life relevance, which many teachers struggle to achieve with traditional methods. By combining automated reuse of materials that the teacher already possesses, generated narratives, and an immersive virtual world, it minimizes teacher effort while providing an engaging learning experience that motivates students. The product vision is summarized in **Error! Reference source not found.**.

Commented [FD34]: Denk dat dit een belangrijke recommendation is maar die volgt niet echt uit feedback van de prototypes. Vermeld ik dit hier zo, vul ik 'teacher feedback' bij prototype 3 aan met feedback die eigenlijk algemeen gegeven werd, of laat ik deze zin hier weg?

Commented [SS34R2]: Ik denk dat dit een belangrijke recommendation is en dan mss best indd toevoegen in prototype?

Commented [SS34R3]: Allez tenzij dat dat echt nooit is vermeld door een van de leerkrachten, maar ik vermoed dat leerkrachten dit ook wel belangrijk vinden?

Commented [AL35]: Not sure if I understood their comment, but got rid of a couple titles. Maybe it needs to be unified a bit more, more overarching?

Commented [SS35R2]: I think they want you to put everything into their template: https://docs.google.com/presentation/d/1ebS-tqBWpOcZEsnQ-SX3atKbrg_Mb5-0/edit?slide=id.p1#slide=id.p1
And then explain rather brief using the filled in template.

Commented [SS35R3]: @Adrien Van Lancker just saw on Teams that they do indeed want you to use their template.

Product vision board

Project name	Group	
Three Point Crown	3	Iteration 5

Product statement Make STEM education tangible and enjoyable through gamified real-life impact representation.

Crisp and granny proof summary of the product / idea

Target group	Needs	Product	Value
Who do you target with your product / idea? Which market segments?	Which problem are you solving for the target group? Why would the target group care?	What are the 3 to 5 top features? Why is your solution unique?	What is the value of your product for the target group?
Primary: STEM teachers in Flemish secondary education. Expected to convey the real-life impact of their subject but often struggle to do so. **Secondary:** Students in their 1st and 2nd year of secondary education. The direct benificiaries of effective, engaging, and enjoyable educational tools. **Tertiary:** Schools, publishers, parents, and policymakers.	**The solution needs to:** • Link STEM subjects to their real-world impact and relevance • Support a variety of STEM subjects • Create engaging learning experiences, through interactive narratives and effects • Maximize material reuse by extracting and enhancing exercises from any PDF • Introduce gamification to make learning enjoyable and motivating • Be easy to use and seamlessly integrate into existing teaching workflows	**Top features:** • Automated exercise extraction and enhancement: Automatically extracts exercises and answers from uploaded PDFs and prepares them for engaging experiences. • Generated narratives for real-world connection: Generates engaging, story-based narratives and effects around collections of exercises. • Engaging gamified virtual world: Students enter an immersive virtual-world where exercises become interactively interwoven with the virtual world's narrative. • Teacher dashboard for monitoring and feedback: Easy-to-use interface to register exercises, track student progress, review and edit generated narratives, and correct/add feedback.	**Primary:** The solution helps Flemish STEM teachers to meet the challenging curriculum objective of linking subjects with their real-life impact, focusing on seamless integration, ease-of-use, and minimal effort. **Secondary:** For students, it helps by making STEM feel more tangible, relevant, and enjoyable through immersive narratives and gamification, leading to improved motivation, interest, and engagement.

Competitors & alternatives

Where are your product's main competitors? Which alternatives is the target group using now?

No tool fully combines automated PDF exercise extraction with generated narratives to link STEM to real-life impact in a gamified context aligned with the Flemish curricula. Alternatives fall short: (i) quiz platforms (e.g., BookWidgets, Kahoot!) lack relations to real-world impact or narrative depth; (ii) simulations (e.g., PhET, MissionUS) offer limited custom exercise support; (iii) interactive tools (e.g., Nearpod) miss automated material reuse, immersion, or low-effort workflows.

2.8 Conclusion of Design Thinking

The Design Thinking phase provided a structured, human-centered approach to exploring and addressing the challenge of connecting STEM education to real-world contexts. This process allowed the team to iteratively refine the project's scope, target audience, and solution direction. Early explorations of different concepts, such as an AI data labeling tool and a history-themed game, initially seemed promising but were ultimately found to address no pressing need.

Through the Empathize Phase, in-depth research with students, teachers, school principals, and educational publishers revealed key needs including engagement, curriculum alignment, ease of use, and progress tracking. These findings were reinforced by the State-of-the-Art analysis, which highlighted that effective digital learning tools combine interactivity, clear curriculum alignment, narrative-driven gameplay, meaningful personalization, and real-world relevance. The SOTA research also emphasized the importance of teacher usability, optimal challenge levels for students, rewards and customization, and progress tracking as critical factors for engagement and learning outcomes.

These insights ensured that the project remained grounded in the real experiences and expectations of its stakeholders. At several points ideas were formed that simply combined aspects of existing tools into one interface. Although this initially appeared practical, it did not solve a new or specific problem. Only through direct interaction with students and teachers did it become clear what challenges genuinely needed to be addressed. This highlighted that assumptions can be misleading and that it is necessary to return to stakeholder feedback during every step of the process to ensure that the solution remains relevant and valuable.

Commented [SS36]: TODO: make this shorter and more summarized

Commented [SS36R2]: Also here is the place to mention our other 'omzwervingen' as Stephanie calls them

Commented [SS37]: Updated this, does it seem okay?

Commented [FD37R2]: Dit is duidelijk en mooi geschreven :-)

Commented [FD38]: @Saira Snauwaert Zou "different concepts" zetten hier, en mss de zin eindigen met "were ultimately found to address no pressing need." omdat de stakeholders nu anders zijn dan de stakeholders toen en ik bang ben dat ze in de war gaan geraken

The Define Phase distilled these insights into the problem statement: “How might we help secondary school teachers link their course material to real-world applications in order to comply with their curriculum objectives and show their students the usefulness of their subject?” This provided a clear focus for developing solutions.

During the Ideate Phase, the team explored multiple concepts, benchmarked existing solutions, and developed the idea of a game-based platform that transforms teacher-provided exercises into contextualized quests for students. Feedback from an educational publisher validated the target audience and highlighted considerations for AI integration, gamification, and progress tracking.

The Prototype Phase enabled iterative testing and refinement, incorporating feedback from students and teachers to improve usability, visual coherence, gameplay mechanics, and learning outcomes. The Test Phase produced functional and non-functional recommendations, which were incorporated into the Product Vision Board to clarify the target groups, value proposition, and differentiating factors.

Overall, the Design Thinking process ensured a deep understanding of stakeholder needs, reinforced by SOTA research, and resulted in a well-defined, user-centered solution that is both innovative and feasible. The process laid a solid foundation for subsequent Requirements and Design phases, ensuring the final product effectively engages students while supporting teachers in achieving curriculum objectives.

3 STATE-OF-THE-ART ANALYSIS

Commented [SS39]: TODO: add conclusions per section or at the end

Commented [SS40]: Is this the correct location for this chapter? I feel like it should be further up in the document.

This chapter presents the results of the State-of-the-Art analysis conducted for this project. Its purpose is to establish a clear understanding of the current landscape in which the serious educational game will be developed. To achieve this, the chapter first brings together insights from multiple domains relevant to the project. In addition, it examines the competitive environment, intellectual property considerations, supporting technologies, market potential, and other findings that may be of relevance to the project.

3.1 General Research

In order to design a serious game that effectively engages secondary school students while helping teachers by linking course content to real-world applications, it is essential to understand the current research landscape on educational tools, serious games, and student engagement. The main areas covered in our research are: digital tools in education, serious games (in educational context), influence of motivation and interest on learning, personalization, and the connection of learning material to real-world applications. The outcomes of this research together with the key take-aways for this project are listed below.

Commented [AL41]: A bit too repetitive with the introduction. Maybe remove the list from the introduction and just use "... from multiple domains relevant to the project." in the introduction?

3.1.1 Digital Tools in Education

Digital tools are increasingly shaping modern learning environments, offering new avenues for engagement and interaction. Students generally perceive interactive tools such as polling apps or gamified quizzes as highly engaging, while passive tools like PowerPoint-only instructions are rated less effective [2]. For this project, this highlights the importance of creating interactive elements that actively involve students, rather than relying solely on passive instruction.

Commented [AL42]: informal: for this, for the, for the proposed solution, ...

From the teachers' perspective, digital tools also play an important role. Many teachers already use technology regularly in their classrooms and during lesson preparation, yet a significant number feel that their students are more proficient with new technologies than they are themselves [3]. Despite this, teachers generally view ICT-based teaching positively, seeing it as a way to make learning more interactive, exciting, and motivating. However, insufficient training or support can hinder effective integration of digital tools in practice [4].

It follows that the tool developed in this project should feature interactive digital elements that actively engage students while remaining intuitive and easy for teachers to use without advanced technical skills. Providing clear guidance, simple interfaces, and optional teacher support materials will help ensure successful adoption and effective use.

3.1.2 Serious Games in Education

Serious games have emerged as a key educational approach that can enhance learning outcomes and engagement when well-designed [5], [6], [7]. Research indicates that gamifying learning content generally increases enjoyment for students [5], [7]. However, the positive effects on knowledge retention are not uniform. Some studies show that gamification does not harm retention but does not necessarily enhance it either [5]. Others suggest that gamification can serve as a powerful tool for

improving student outcomes if it is thoughtfully designed and closely aligned with educational objectives [6]. For this project this implies the following: the game should be designed with clear educational objectives in mind. Gamification can make learning more enjoyable, but we cannot rely on it alone to improve retention, content alignment is of high importance.

Providing clear guidance to students playing the game (showing them objectives, rules, and expected outcomes) is another critical factor for effectiveness [8], [9]. Games that are explicitly tied to the curriculum are more likely to be accepted and integrated into classroom programs by teachers [8], i.e. there should be a balance between play and real-world learning [8], [9]. Equally important is calibrating challenge levels: tasks should be neither too difficult, which risks demotivating players, nor too easy, which can make the game feel uninteresting. Optimal challenge levels are key to sustaining engagement [8], [9], and care must be taken to avoid cognitive overload [10]. The tool created in this project must thus provide clear instructions and objectives, integrate curriculum-aligned content, and maintain motivation without overwhelming players.

3.1.3 Promoting Student Interest and Motivation

Student interest and motivation is a major influence of learning outcomes [11], [12]. When students are genuinely interested in a topic, they are more likely to engage deeply with the learning material and retain information [11]. Gamified learning environments can further enhance motivation, as learners often respond more positively to game-based methods than to traditional instruction [7]. Narrative-driven games, in particular, have been shown to boost engagement, foster empathy, and improve learning outcomes. Incorporating meaningful stories, relatable characters, and immersive scenarios makes educational experiences more memorable and motivating [9], [13], [14]. In order to maximize engagement and learning, the game developed in this project should include narratives and gameplay that spark students' interests. Integrating story elements and relatable characters will help sustain motivation while also deepening understanding.

3.1.4 Personalization

The role of personalization in educational games is nuanced. Some studies report no significant effects of personalization on motivation, enjoyment, or flow, while others find that well-designed customization can reliably increase motivation and related outcomes, which may indirectly support learning performance [15], [16]. However, early and thoughtful personalization using contextual information such as school profiles or social media data can support tailored learning experiences [17]. The key-takeaway for the project is that while basic personalization may not always enhance engagement, more targeted approaches that align with student context and are meaningfully integrated into the game's experience could still be valuable.

3.1.5 Real-World Application

Research shows that secondary school students demonstrate increased motivation, engagement, and conceptual understanding when lessons are connected to authentic, real-world problems [18], [19]. Problem-based teaching approaches and emphasizing relevance of the taught material are said to spark situational interest in learners [11]. Moreover, integrating practical problem-solving and real-life applications into gameplay can help students see the value of their learning and better retain the knowledge, while simultaneously fostering skills like critical thinking and creativity [9].

Commented [Adrien Va43]: I would add another paper as there are studies that show that customization/personalization consistently improve motivation, engagement, etc. which in turn can improve learning performance. It is rather difficult because as far as I can tell the performance impact is indirect customization -> may improve property x -> may improve learning objectives. Also another thing I conclude from this nuance is that impact of customization/personalization strongly depends on whether it is well designed, i.e. does it add to the game experience compared to if it was not added or does it take away from it.

Another decently cited paper which supports customization: https://scholar.google.com/scholar?hl=en&as_sdt=0%2C5&q=The+effects+of+customization+on+motivation+in+an+extended+study+with+a+massively+multiplayer+online+roleplaying+game&btnG=

Commented [SS43R2]: I don't completely understand what you mean, feel free to add the info to the paragraph. I can then add the reference if this doesn't work in your online Word.

Commented [AL44]: Lichtjes aangepast.

Commented [SS44R2]: Kun je even ergens anders iets typen, ik kan de ref niet toevoegen omdat jij de laatste aanpassing hebt gedaan

Commented [AL45]: @Saira Snauwaert Reference: https://scholar.google.com/scholar?hl=en&as_sdt=0%2C5&q=The+effects+of+customization+on+motivation+in+an+extended+study+with+a+massively+multiplayer+online+roleplaying+game&btnG=

This focus on real-world relevance aligns strongly with the Flemish curriculum for the first grade in the A-stream. Several curriculum objectives explicitly emphasize linking learning content to society. In particular, objective 06.44 states that *"students illustrate the interaction between science, technology, mathematics, and society using societal challenges* [20]*."*

This implies that the tool developed within this project should incorporate real-world challenges that clearly demonstrate the interaction between STEM subjects and society. Doing so will not only enhance student motivation and conceptual understanding but will also align with Flemish curriculum objectives, increasing the likelihood that teachers will adopt and integrate the game into their lessons.

Commented [Adrien Va46]: informal

3.1.6 Key Takeaways from General Research

Based on the reviewed literature and insights from educational research, the following key points are relevant for the development of the gamified educational platform in this project:

- **Interactivity increases engagement:** Students respond more positively to interactive elements, while passive tools are less effective. The platform should therefore prioritize interactivity to maintain engagement.
- **Usability for end users is crucial:** The system should be simple, clear, and easy to adapt for both students and teachers.
- **Curriculum alignment is essential:** Serious games are more effective when they have explicit educational objectives and are tied to the curriculum. Clear instructions, objectives, and curriculum alignment should be integrated into the game.
- **Narrative and storytelling enhance learning:** Relatable characters, immersive scenarios, and narrative-driven gameplay increase motivation, engagement, and retention. Storytelling should be a core element of the game design.
- **Personalization can support learning:** Thoughtful, contextually meaningful personalization may enhance motivation and engagement.
- **Real-world relevance boosts engagement:** Linking exercises and challenges to real-world applications helps students understand the value of what they are learning, improves conceptual understanding, and aligns with Flemish curriculum objectives.
- **Progress tracking is important:** Both students and teachers benefit from timely feedback. The platform should allow students to see the consequences of their actions (e.g., through visual effects in the game) and provide teachers with a clear overview of student progress.

Commented [FD47]: Zou dit verkorten tot: "Students respond more positively to interactive elements than passive tools." om minder in herhaling te vallen

Commented [SS47R2]: Mathias wou de conclusies kunnen lezen zonder de rest van de SOTA te lezen dus vermoed dat dit er wel in moet. Ik besef dat het immens herhalend is maar ja...

Commented [SS47R3]: Heb de voorbeelden eruit gehaald, heb beslist als ze meer info willen dat ze maar alles moeten lezen.

3.2 Competitive Analysis

This section examines existing educational software and serious games to identify current approaches, strengths, and limitations. Understanding the competitive landscape helps determine opportunities for differentiation and guides the design of the project's game-based platform. By reviewing interactive simulations, gamified assessment tools, teacher support systems, and engagement strategies, the analysis highlights how the proposed solution can address gaps and improve upon existing offerings.

3.2.1 Competitive Landscape

A review of current educational software and serious games reveals a variety of approaches, each showing both strengths and limitations. Many platforms focus on interactive simulations, particularly in STEM subjects. For example, PhET provides students with exercises in physics, chemistry, biology, and mathematics through simulations, allowing them to explore concepts dynamically and gain hands-on experience [21]. Similarly, curated lists of top educational software include subject-specific interactive simulations, such as virtual physics labs or mathematical modeling tools [22]. These tools are effective at making abstract concepts tangible, but they are often limited to specific domains. This project could differentiate by adding narrative or problem-solving contexts and by covering the different STEM subjects.

Quiz-based and gamified assessment platforms are also prevalent. Platforms such as Kahoot! [23] and Blooket [24] offer competitive, multiplayer quizzes with customizable content, fostering engagement and motivation. These platforms are particularly effective for immediate feedback and revision, but they typically emphasize recall over deep conceptual understanding. While quiz-based platforms excel at engagement and quick feedback, the game created in this project can differentiate by integrating deeper learning experiences, problem-solving, and narrative-driven challenges alongside gamified assessment. By combining motivation with meaningful cognitive tasks, the game can foster both engagement and conceptual understanding, rather than focusing primarily on recall.

Other platforms focus on simplifying teacher workflows and integrating AI tools. For example, PDF-to-Quiz [25], GetMarked [26], and QuizGecko [27] allow educators to convert existing materials, such as PDFs or Word documents, into interactive quizzes. These tools demonstrate the potential of AI to reduce preparation time and support curriculum integration, highlighting the importance of focusing on the ease of use for teachers.

Several platforms also integrate rewards and customization to motivate students. Bingel [28] and Blooket [24] show the effectiveness of allowing students to personalize avatars, while PhET and other simulation platforms use point systems and interactive progress tracking [21], [24], [28]. This demonstrates that combining engagement mechanics with learning content can enhance motivation, while still supporting curriculum goals. To boost motivation and engagement, the game created in this project could also implement reward systems and customization options, such as points, badges, or avatar personalization, allowing students to track progress and feel ownership over their learning experience.

3.2.2 Key Takeaways from Competitive Analysis

The competitive analysis highlights key insights for designing an effective game-based learning platform:

- **Interactivity increases engagement:** Active participation is crucial. The platform should incorporate gameplay, problem-solving, and other interactive components to maintain student engagement.
- **Usability for teachers is crucial:** The system must be intuitive and easy to use, allowing teachers to efficiently create, assign, and track exercises while integrating seamlessly into their existing workflow.

- **Rewards and customization boost engagement:** Motivation can be enhanced through meaningful rewards and personalization, enabling students to track progress and feel ownership over their learning.
- **Narrative and storytelling enhance learning:** Embedding exercises within short, context-rich stories helps students understand the relevance of what they are learning and strengthens the connection between STEM concepts and real-world applications.

3.3 Supporting Technology

This section discusses supporting technologies that are relevant to the development of the proposed solution. The goal is not to make concrete design decisions, but to outline technological directions and constraints that influence later architectural choices.

3.3.1 Technology Landscape

Generative AI tools like ChatGPT [] and Microsoft Copilot [] are increasingly used by students []. These tools could support the project in several ways, for example by helping generate content, providing hints, or creating educational scenarios. However, their use in an educational context must be carefully designed to ensure that they support learning goals rather than replace critical thinking or teacher involvement.

Game engines such as Unity [] and Unreal Engine [] represent mature platforms for developing interactive and cross-platform educational experiences. Unity is commonly used for both 2D and 3D educational games due to its versatility, extensive tooling ecosystem, and support for deployment on a wide range of devices []. Unreal Engine provides advanced graphics and physics capabilities, making it suitable for highly immersive 3D environments [].

Both engines support rapid prototyping, integration of multimedia content, and interactive features. As such, they illustrate the types of technologies that can enable the development of engaging, game-based educational experiences.
The implications of these technologies for architectural design and feasibility are further evaluated in Chapter 7.

3.3.2 Key Takeaways from Supporting Technology

The supporting technology landscape highlights several constraints and opportunities relevant to this project:

- AI technologies offer strong potential for content generation and automation but require careful integration to preserve educational value and teacher control.
- Processing educational content locally is preferable to external APIs due to privacy and intellectual property considerations.
- Game engines provide a proven foundation for interactive and narrative-driven learning environments, but technology choice must balance realism, complexity, and development feasibility.
- Event-driven and modular architectures are well suited for educational games that combine user interaction, learning feedback, and visual effects.

These insights directly inform the architectural and technology decisions discussed in Chapter 7.

Commented [SS48]: Eigenlijk is het verplicht, maar de Act is nog niet van toepassing denk ik dus smijt ik het er wel uit hahah

Commented [SS49]: @Francisco Van Langenhove please add to this and then write a conclusion or refer to your technology chapter that the takeaways are discussed there. Or do it the other way around.

Commented [AL50]: They are game engines, exploration engines are things you can build with them.

Commented [SS50R2]: Are they Immersive game-engines? Or is just game-engines fine? This is just info I found on Internet haha

Commented [AL50R3]: Just game engines (will never be wrong).

3.4 Intellectual Property and Rights

When developing the serious educational game, it is essential to ensure compliance with both copyright regulations and legal frameworks governing data protection and the use of artificial intelligence. Two major areas are particularly relevant for this project: protection of user data under the General Data Protection Regulation (GDPR) and compliance with the requirements of the EU AI Act.

The GDPR establishes strict rules around the collection, processing, and storage of personal data []. For this project, this means that users must be clearly informed about what data is being collected and why, and the amount of personal data processed should be kept to an absolute minimum. Students and teachers must have access to mechanisms that allow them to download, review, or delete their data at any time. Because the tool will also be used by minors, special attention must be paid to transparency, parental or institutional oversight when required, and secure data management.

In addition to GDPR, the newly approved EU AI Act [] introduces an additional regulatory layer relevant to systems that use artificial intelligence. The Act classifies AI systems into risk categories and sets corresponding obligations for developers and deployers. Because in this project AI will not be used to evaluate students, the platform would not fall under the category of high-risk AI systems. It would be a so-called limited-risk AI system. These systems are described as being able to pose some risk to users or consumers, e.g., generate or manipulate content or provide chatbot services []. This means in this project, care must be taken to provide users with clear information about their nature and purpose. Content that was generated by AI must be labelled accordingly [].

3.5 Target Group and Market Potential

The target market for this serious educational game primarily consists of secondary school students, their teachers, and, less directly, principals and parents. Several sources indicate that the demand for digital and gamified learning solutions is growing, highlighting both market opportunity and potential user interest.

3.5.1 Market Analysis

The global K-12 online education market is expanding rapidly, driven by rising government investments in digital infrastructure and growing demand for personalized learning [36]. In 2024, the market was valued at USD 171.5 billion and is projected to reach USD 228.27 billion by 2033. Laptops and Chromebooks are the most commonly used platforms, and many schools adopted online learning frameworks quickly during the pandemic [36]. This suggests that secondary school students are increasingly comfortable with digital learning tools, creating a favorable environment for introducing a digital educational game.

In addition to overall digital education growth, the demand for game-based learning is also increasing. The game-based learning market is growing rapidly, primarily driven by demand for personalized and adaptive learning solutions [37]. In Europe, the serious games market is expanding, with increasing adoption of gamification in education; it is estimated that 70% of educational institutions will adopt some form of game-based learning in the future [38]. These trends suggest that there is a large and receptive audience for engaging, gamified educational tools, highlighting the potential for this project to meet both student and institutional needs.

Commented [FD51]: Voorstel: "When developing the serious educational game, it is essential to ensure compliance with both copyright regulations and legal frameworks. Saving data about students, teachers, and exercises is necessary for the working of our platform. That means that the relevant data protection laws must be known and followed. Our platform will make use of AI agents to implement certain functions, as is explained in the technology chapter. Because of that, we need to comply with the laws about the use of artificial intelligence. Two major areas are particularly relevant for this project: protection of user data under the General Data Protection Regulation (GDPR) and compliance with the requirements of the EU AI Act."

Commented [FD51R2]: Reden voor voorstel: hiervoor wordt niet echt over AI in context van ons project gesproken en anders komt het precies uit de lucht vallen

Commented [FD51R3]: We zouden ook kunnen 3.3 en 3.4 omwisselen om al gezegd te hebben "strong promise in AI tech" voordat we over AI wetten beginnen

Commented [SS51R4]: Zal ze omwisselen

Commented [AL52]: Immaculate

Commented [FD53]: Hier ook onze eigen enquête vermelden over dat leerkrachten het zien zitten om AI te gebruiken (want das echt letterlijk market analysis).

Commented [FD53R2]: + enquête dat leerlingen willen lopen?

Commented [FD54]: Zou hier serious vervangen door digital

As students in secondary school are too young to pay for the platform themselves, either the principal of a school would have to pay or the parents of the students. The principal's needs and willingness to pay was analyzed in the empathize phase and can be found in the principal persona.
Concerning parents' willingness to pay, the following was found: Parents' willingness to pay for online learning solutions depends strongly on perceived value, particularly whether the platform is clearly linked to school courses and demonstrates practical usefulness [39]. This implies that adoption of the serious game will depend on clear alignment with curriculum objectives and demonstrable learning benefits.

3.5.2 Key Takeaways from Market Analysis

The analysis of the target market and potential demand indicates a strong opportunity for the proposed serious educational game. Secondary school students are increasingly familiar and comfortable with digital learning tools, creating a favorable environment for introducing interactive and gamified solutions. Growth in both the global K-12 online education market and the game-based learning sector suggests a rising interest in personalized, adaptive, and engaging educational tools. Successful adoption will depend on clearly demonstrating educational value, aligning exercises with curriculum objectives, and showing practical benefits for both students and teachers. Parents' willingness to support or invest in such solutions is closely linked to perceived relevance and usefulness, emphasizing the importance of a well-integrated, curriculum-aligned platform. Overall, there is significant potential for a gamified educational tool that combines motivation, learning effectiveness, and usability for teachers.

Commented [AS55]: Secondary school students?

3.6 Other Findings

Commented [SS56]: Doesn't necessarily need to be called Other Findings

Commented [SS56R2]: Also does anyone know any other things to add here? Maybe some technology research?

In addition to research on serious games, digital tools, and market trends, it is important to consider how to evaluate the game once developed. Several validated rating scales exist for assessing game experience, each with a slightly different focus.

GUESS18 [40] is the shortened version of the Game User Experience Satisfaction Scale (GUESS). It is primarily used to determine overall satisfaction with a game, making it highly suitable for measuring general "fun" or enjoyability. While it is not specifically designed for educational games, its brevity and broad focus on enjoyment make it particularly useful for our project, given that enjoyability is a key quality attribute.

EGameFlow [41] is designed specifically for e-Learning games. It measures enjoyment in the context of learning, considering factors such as immersion, challenge, feedback, and perceived knowledge improvement. This scale is relevant if the game explicitly ties gameplay to educational objectives and if the connection between fun and learning is a focus.

The Game Experience Questionnaire (GEQ) [42] captures more nuanced aspects of player experience, including immersion, flow, competence, affect, challenge, tension, and optionally social presence. It provides a comprehensive view of user experience beyond simple enjoyment.

MEEGA+ [43] is a method and questionnaire for assessing the quality of educational games, evaluating both usability and learning experience. Developed and validated in an educational context, it is particularly useful for serious games aimed at school environments.

For this project, GUESS-18 [40] appears to be the most relevant scale, since the primary quality attribute that will be measured using a rating scale is the enjoyability of the game (see Chapter 5).

Commented [SS57]: TODO: reference the requirements chapter properly

Using this scale will assist in evaluating the degree to which the game successfully engages students and provides an enjoyable learning experience.

3.7 Conclusions of SOTA

Commented [SS58]: Ik besef dat ik immens in herhaling val in dit hoofdstuk, maar ze vroegen conclusies dus ik weet niet echt of dit erg is dan.

Commented [FD58R2]: Niet erg, mooi geschreven conclusies.

The State-of-the-Art analysis provides a comprehensive overview of the landscape in which the serious educational game will be developed, highlighting key insights from research, competitors, technologies, and market potential. The analysis confirms that interactive, narrative-driven, and contextually meaningful games can increase student engagement, motivation, and understanding, particularly when aligned with curriculum objectives and real-world applications. Teacher usability and workflow integration are critical factors for adoption, while rewards, customization, and clear feedback mechanisms enhance both motivation and learning outcomes.

Market trends show strong potential for gamified educational tools, with secondary school students increasingly familiar with digital learning platforms and institutions showing growing interest in game-based solutions. Parental support is influenced by perceived educational value, emphasizing the need for clear alignment with school curricula.

Competitive analysis demonstrates that while existing platforms provide interactivity, gamified quizzes, and teacher support, there is an opportunity to differentiate by combining multiple STEM subjects, integrating narratives, and balancing engagement with deeper learning.

From a technological perspective, the State-of-the-Art analysis indicates that modern educational platforms increasingly rely on a combination of AI-driven content generation, interactive game engines, and modular system architectures. Existing tools highlight the potential of generative AI to reduce teacher workload and support content creation, but they also expose important constraints related to privacy, intellectual property, and the controllability of generated output. In addition, current serious games show that scalable, event-driven architectures and lightweight game engines are well suited for interactive learning environments that must operate reliably on heterogeneous school hardware. Together, these insights point towards the need for a technology stack that balances expressiveness, performance, and maintainability, while respecting privacy and deployment constraints.

Overall, the findings underscore the importance of designing a platform that is engaging for students, easy to use for teachers, curriculum-aligned, and capable of providing clear feedback and meaningful learning experiences. The insights from the SOTA analysis will guide the ongoing design and development of the educational game, ensuring that it addresses both educational objectives and user needs.

4 GAME

In this chapter, the most important elements of the game are described. This chapter is meant to give an overview of how the student will experience the platform. It also introduces the terminology needed to understand the following chapters.

In the following sections, a detailed description of the game is provided. The world, the game loop and other related subjects such as player customization are discussed.

4.1 The Game World

The game world consists of different NPCs and structures, which inhabit all the different districts. Both the NPCs and the structures can be influenced by the player's decisions in quests.

4.1.1 Player Perspective and Situated Time

The game will be a 2.5D single player video game. Games which are 2.5D have 2D visual aspects but give the players a feeling of depth with the use of perspective. They allow players to walk 'up-and-down' the screen in addition to 'left-to-right'. This feeling of depth is included to improve the immersion for the students. The choice for this perspective and the choice to allow students freedom of movement to walk around in the world were made after a discussion with 10 students of the 1st grade in secondary school. The player can walk around the game world freely and can interact with the world and the NPCs in it.

The game is situated in the present time. This means the structures and the professions of the NPCs can be modern without the restriction of a certain time period. Since the goal is to connect learning material to the real world, it's important that real present time context can be used in the game.

4.1.2 Districts

The world is divided into different districts, one for every STEM subject. The number of districts a student sees depends on how many of their STEM teachers use this platform. The reason the world is divided into districts is to give student a good overview of their subjects, as we noticed that they found this very important during the testing of our first two prototypes. Each subject's effects stay contained within its district.

In our world, every district has one main street running from the left to the right. From this street, side streets diverge which go up and down the screen. The reason to have the main street running horizontally is to properly fill up a modern computer screen, which is wider than it is high. The side streets on the other hand will make the world feel more lively and almost three dimensional.

4.1.3 NPCs

The player interacts with NPCs to complete quests. Each NPC has a profession. For example, an NPC might be a chemist, a hunter, an architect, … The professions included in the game are gathered from a questionnaire that 16 teachers filled in. In this questionnaire we asked them what professions are most closely linked to their subject material and which jobs they would like

Commented [SS59]: Is het een idee om dit voor de requirements phase te zetten? Want daar doen we een hele introductie van terminology die ik eigenlijk beter vind passen in deze sectie.

Commented [SS60]: General comment, I find this section a little bit chaotic, there are many subtitles which only have 1-2 sentence below them. Either the content should be workded out further or there just shouldn't be subsections.

Commented [SS61]: Intro needed that explains what this chapter is about

Commented [FD61R2]: This alright?

Commented [SS62]: Moet dit ook niet in de sectie over de game world? Vind het raar om dit in de intro te zetten van het hoofdstuk

Commented [SS63]: Is this true?

Commented [SS64]: Wouldn't mention this here, this is mentioned in the next chapter

Commented [BR65]: Voor zover ziet het er al goed uit maar persoonlijk zou ik dit weglaten

Commented [FD65R2]: Dat weet ik. Persoonlijk zou ik het laten staan.

Commented [SS66]: Seems like a very broad profession, would split this up in different ones or just mention a different example here.

represented in our platform. We aim to include a wide range of NPCs to allow for a diverse and relevant coverage of professions.

4.1.4 Buildings/Structures

There are different buildings in the game to accompany the story elements. These buildings can improve or deteriorate depending on how players complete quests. Some of these structures will be tied to professions, such as a lab for a chemist or a hospital for a doctor. Other structures are more general such as the city hall and the streets themselves. These can also improve as a result of a quest where a specific NPC decides to undertake a project using their knowledge to improve the city. We will predefine 5 effects per structure to be able to see a gradual decay/improvement. They are meant to situate the question in an applicable context so that the teacher can meet their learning goal related to the real-life context of their material and so that the student can see the relevance of what they learn in the real world.

The Game Loop
NPCs of the activated quest ask the player for advice. Based on the helpfulness of the player's advice the situation and the emotional state of the NPC changes. For example, if an NPC asks a player for financial advice and this advice is correct, this character can afford to buy some new flowers for their garden and becomes happier. Wrong answers can make structures worsen and NPCs sad. For example: students need to use physics to calculate the wind pressure that will be on the windows during a storm. If this pressure is known, adequate measures (such as installing plywood covers or hurricane film) can be taken to prepare. A wrong calculation would cause the windows to break during the storm (if not enough protection) or cause the owner of the building to spend too much money on the protection. However, re-attempting failed quests restores the structures. It is important that the effect makes sense in the context of the attempted exercise.

The player, who is either a prince or a princess, has their own palace in the middle of all the districts. They cannot go inside this palace. With their coins they can upgrade the front view of their palace. If they click on their palace, they can see their progress over all the subjects. Progress is shown both globally and per subject. By globally, we mean e.g. total amount of quests made and total amount of quests solved correctly. Per subject we will also show how many quests were made, how many of them were correct, how many attempts they needed, how much time they have spent practicing…

We consider an exercise to be a question-answer pair extracted from uploaded PDFs without any generated context. From these exercises, our system generates quests. A quest consists of at least one exercise. Each exercise in the quest belongs to the same subject. Together, these exercises receive a generated context. Each exercise is represented through an interaction with an NPC which provides the context along with the exercise question. These interactions all take place in the same district (the district of the subject), and follow the logical order of a story with a coherent narrative. After each interaction, the student gets to see whether they answered correctly and at the end of the quest they can see the full effects of their actions.

When quests are completed correctly, the world improves and students earn coins. These coins can be used to buy cosmetics for their character in a shop or to upgrade their palace.

Commented [BR67]: Dit gaat een beetje off-topic voor het deel dat de game beschrijft, en zou ik weglaten

Commented [FD67R2]: Ik zou het zeker vermelden aangezien ze in het verleden al gevraagd hebben hoe we de professions kiezen en waarop dat gebaseerd is

Commented [BR68]: Dit zou ik in 4.1 The Game World plaatsen, aangezien het beschrijft wat NPCs zijn

Commented [FD68R2]: Ik zag wereld eerder als gebouwen enzo, en NPCs als deel van het spel

Commented [BR68R3]: Ah. Ik had de structuur op deze manier gemaakt met de bedoeling van: Eerst bespreken we welke elementen er allemaal in de game zijn, en daarna leggen we de gameplay uit en hoe de student interageert met de wereld in de game loop

Commented [SS69]: Vind de onderverdeling wat raar, waarom zit oplossen van quests in game world en niet in game loop?

Commented [SS69R2]: En customization zit dan wel in game loop terwijl dat in mijn hoofd gewoon acties zijn op de game world?

Commented [FD69R3]: Ah sorry das afkomstig van de onderverdeling die Briek eerst gemaakt had maar ondertussen heb ik het teveel veranderd denk ik

Commented [SS70]: Feels weird to mention this before quests are ever mentioned

Commented [SS71]: Dit heeft precies weinig te maken met die storm? Zou dit deel van de zin weglaten

Commented [BR72]: Misschien de voorbeelden na de algemene beschrijving van de states en effecten? Om de vlotheid en structuur van de tekst wat te verbeteren?

Commented [FD72R2]: Das goed voor mij.

Commented [BR73]: Ik wist niet dat we dat nog zouden doen? Ik dacht dat we het paleis hadden weggelaten en als het paleis er was, dat er maar 1 paleis was voor alle districts. (zoals in het 3e prototype?)

Commented [FD73R2]: 1 paleis voor alle districts is exact wat ik bedoel. 1 palace in the middle of all the districts (en dus niet in een specifiek district)

Commented [BR73R3]: Aah oké

Commented [BR73R4]: Maar ben nog steeds niet zeker of het paleis echt nog in de game zit. Ook omdat het in deze versie niet kan betreden worden lijkt het mij niet zo nuttig meer

Commented [FD73R5]: We hebben het daar gisteren wel over gehad dat dat een manier kan zijn om studenten naar hun progress te laten gaan.

Commented [BR74]: Was dit al beslist? Ik dacht dat dit maar een suggestie was maar nog niet echt vastgelegd was?

Commented [FD74R2]: We zijn toch absoluut zeker dat de wereld verbeterd als ze dingen juist doen?

Commented [BR74R3]: Aah gewoon als er staat "students earn a better world" lijkt het alsof de studenten een geupgrade nieuwe wereld krijgen ipv dat de wereld verbetert doordat een gebouw in de wereld verbetert

Commented [FD74R4]: Beter?

Commented [SS75]: Dit is ookal gezegd hierboven

When playing the game, the student first selects a specific district. Inside this district they can see their quest list for the relevant subject. When they select a quest from this list, an arrow appears on their screen to guide them towards the location of the NPC who will start this quest. They must walk over to this NPC and then talk to them to get the explanation of the quest. For simple quests, they will only talk to one NPC and complete an exercise for them. For bigger quests combining multiple exercises, students may need to converse with multiple NPCs to be able to complete the quest.

When they are doing a quest and decide that they would rather do something else, they can either exit the district or select a different quest from the quest list. Doing so will discard their progress on the current exercise. If the quest contains multiple exercises, the result of completed exercises is saved. In that case, if they re-attempt the quest, they start from the exercise where they ended last time. The game is unaffected by exercises that are not completed as they are never submitted.

5 REQUIREMENTS

The completion of the Design Thinking phase marks a shift from exploring possibilities to defining what will be developed in practice. In the requirements phase of the project the insights from interviews, the State-of-the-Art analysis, and the prototypes were translated into a set of concrete requirements. Based on these requirements an initial view of the software architecture was outlined that indicates the main components of the system and how they relate to each other. These steps prepare the ground for later implementation.

Commented [SS76]: Delete this section and rewrite this to be introduction to requirements chapter

Commented [FD76R2]: Like this?

Commented [SS76R3]: Yes, looks good to me!

5.1 Use cases

The actors interacting with the system are secondary school STEM teachers, secondary school students, and the school's IT coordinator.

Below the complete list of use cases of the product is presented. The different columns indicate the number, name, involved actors, priority, complexity and frequency of the use cases. Below the table a more detailed explanation of all the use cases in the list is given.

Use Case List

#	Description	Frequency	Priority	Complexity
1	As a **teacher**, I can upload a pdf from which the system extracts exercises pairs.	Once every few weeks	High	High
2	As a **teacher,** I can generate real-life context for my exercises.	Once every few weeks	High	High
3	As a **teacher**, I can edit the generated real-life context.	Once every few weeks	High	Low
4	As a **teacher**, I can edit the automatic correction of an exercise.	Once every week	Medium	Low
5	As a **teacher**, I can combine multiple exercises into one quest.	Once every few weeks	Medium	Low
6	As a **teacher**, I can reuse exercises/quests across classes.	Once every few weeks	Medium	Low
7	As a **teacher**, I can share exercises/quests with other teachers.	Every couple months	Low	Low
8	As a **teacher**, I can check the progress of my students individually.	Once every few days	High	Low
9	As a **teacher**, I can check the progress of my students per class.	Once every few days	High	Low
10	As a **teacher**, I can provide general feedback to every student who gets a certain quest wrong.	A couple of times a week	Low	Low
11	As a **student**, I can make my character.	Once, when account is created	Low	Low
12	As a **student**, I can customize my character.	Multiple times / day	Low	Low

Commented [SS77]: Stephanie zegt: ik zou bij de top 3 nog een use case toevoegen omtrent feedback/evaluatie van oefeningen vanuit perspectief teacher. Gezien die 13/11 wel bijna alles langs kant van student omvat

Begrijp niet wat ze bedoelt maar is dit een goeie voor de top 3?

Commented [AS77R2]: Ik ben akkoord! Denk dat dit een feature zal zijn waar de leerkrachten veel waarde aan hechten en zo kunnen ze manuele controle behouden

Commented [SS78]: Das wel veel toch, maar de commentaar was dat "once every few weeks" te weinig was

Commented [FD78R2]: Once every week?

Commented [AS79]: Extra use case die ik heb toegevoegd omdat die nodig was bij hoe we een "exersise" zien. (zie comment bij de terminology list)

Commented [SS79R2]: Complexity dan High? omdat we met regeneraten zitten en alles met generating context in de use case heeft high complexity gekregen

Commented [SS79R3]: Andere verwoordingssuggestie: "I can combine multiple exercises into 1 quest"

Commented [FD79R4]: Ik ben akkoord met Saira dat het beter klinkt als we zeggen "I can combine multiple exercises into one quest."

13	As a **student**, I can roam around in the game world.	Multiple times / day	Medium	Medium
14	As a **student**, I can solve quests by talking to NPCs and see the impact of my subjects on the real world through visual effects happening to the corresponding district.	Multiple times / day	High	High
15	As a **student**, I can remake quests.	Multiple times / day	Medium	Medium
16	As a **student**, I can check my progress per subject.	Multiple times / day	Medium	Low
17	As the **IT-coordinator**, I can create an instance for the school.	Once	High	Low
18	As the **IT-coordinator**, I can create teacher and student accounts for my school.	Once	High	Low
19	As the **IT-coordinator**, I can configure the class groups at the beginning of the school year.	Once	High	Low
20	As the **IT-coordinator**, I can assign subjects and teachers to classes.	At the start of every school year	High	Low
21	As the **IT-coordinator**, I can edit the class configurations.	Whenever a student changes classes	High	Low

Commented [FD80]: an instance is zo vaag. an instance of what? mss beter "an environment" ?

Commented [SS80R2]: Had de formulering overgenomen uit de commentaar uit de eerste versie

Commented [AS81]: Deze zin klinkt nie echt, maar ik krijg het precies ook nie beter

Commented [SS81R2]: "I can assign subjects and teachers to classes at the start of the school year." Zoiets?

Commented [SS82]: Misschien is de "at the start of the school year" zelfs overbodig

Use case 1: As a teacher, I can upload a pdf from which the system extracts exercises pairs. The teacher needs to be able to upload exercises in pdf format. These documents can vary from self-made exercises to the exercises a teacher receives from the publisher of the schoolbook they are working with. A teacher can upload exercises as many times as necessary. This means they only have to use this feature whenever they have new material. For experienced teachers with a lot of exercises this will most likely only be once, for newer teachers this could be every few weeks. The extraction of the exercises needs to be as faultless as possible, because the teachers should not have to work extra to correct wrong exercises. Because high accuracy is so important here, this use case was granted a high complexity label. Extracting STEM exercises and solutions from a pdf document will most likely prove to be a challenge since they often have complex layout or symbols. This use case is one of the most important pillars of the project and thus has a high priority.

Commented [SS83]: Zou hier tussentitels van maken, ze hoeven niet per se een nummer te hebben maar als je bv de use case in het vet zet en dan de uitleg doet denk ik dat het overzicht beter bewaard zal worden.

Use case 2: As a teacher, I can generate real-life context for my exercises. This is another key aspect of the project, as linking the exercises to a real-world context is the main purpose of our platform. Therefore it also received a high priority label. This generation needs to happen whenever teachers make a new quest or whenever they convert all exercises extracted from a document to individual quests. We believe this will mean once every few weeks. Because generation does not have a correct answer but still has to be of certain quality, this aspect will require quite some experimentation and trial and error. For that reason it is marked as high complexity.

Use case 3: As a teacher, I can edit the generated real-life context. When a context is not up to par with the teacher's standards or expectations they need to be able to edit it. This editing will only need to happen whenever the teacher is making new quests and should not take too long. This should not be too complex to implement. For more in-depth edits, the teacher can select NPC's

themselves and rewrite their dialogue. There will also be an option to provide general feedback on the generation for re-generation. This should be relatively straightforward when generation is already implemented. This use case has a high priority as it makes the platform a lot more user-friendly for the teacher. We consider reviewing the context (even if no changes are made) part of this use case.

Use case 4: As a teacher, I can edit the automatic correction of an exercise. When a fault is discovered in one of the uploaded pdfs, either because there was a mistake in the uploaded pdf or because our system did not parse the pdf correctly, it must be possible to correct it. As changing the answer does not change the context, this is not complex to implement. Ideally this scenario does not occur often, but should it appear, it is a necessary feature. This means this use case has a medium priority.

Use case 5: As a teacher, I can combine multiple exercises into one quest. A quest consists of at least one exercise. Sometimes, multiple exercises build on each other or are closely linked. In that case, the teacher should be able to combine these exercises into a singular quest. This also means that a context has to be generated for these exercises together. This should not be complex to implement because it will make use of the same generation mechanism that is used for one exercise. It will most likely only be needed every time a teacher has new material to upload. This use case receives medium priority. It is important, but the initial focus is on getting the product to work for quests consisting of one exercise.

Use case 6: As a teacher, I can reuse exercises/quests across classes. Teachers need to be able to use the quest for multiple classes if they want to, without having to repeat the upload- and generation process again. When the teacher makes new quests, they should be able to select which classes get these quests. This is low in complexity. The priority is medium as the absence of this feature would cause a higher workload for the teachers.

Use case 7: As a teacher, I can share exercises/quests with other teachers. Teachers should be able to share their exercises with their colleagues if they wish to do so. From teacher interviews it was concluded that teachers who work at the same school sometimes share material. This normally is an event that does not occur a lot, most likely only once every couple of months. This is medium priority. It is not essential for the platform to be useful but is nice to have. This should not be complex to implement.

Use case 8: As a teacher, I can check the progress of my students individually. As students are completing (or failing) quests, it is important that a teacher can look at their progress individually. This is one of the main things teachers want from the platform and therefore it has high priority. Teachers should be able to see how much a student has practiced, their success rate, and how long the student has spent on exercises. They will only see the progress for the students that they themselves teach. This is not a complex aspect, but brings a lot of value for the teacher as they can use this data to decide what their best next course of action should be. Teachers should be able to check this as much as they want to, which we expect to be every week.

Use case 9: As a teacher, I can check the progress of my students per class. It is important that a teacher has a progress overview of the entire class. This progress is shown in the form of a list overview per quest that tells the teacher how many students have started a quest and how many of them have successfully completed it. This way they can estimate the class's skill level and look if there is a subject that needs more attention in a certain class. This is low complexity to implement and

Commented [SS84]: Maybe mention why?

Commented [SS84R2]: Is it because if generation on its own is already implemented, regeneration cannot be too complex to implement? And editing?

Commented [AS84R3]: Ik heb het hier toegevoegd en bij editing ook, is het zo duidelijker?

Commented [SS84R4]: Yes, ziet er goed uit!

high priority as this is important to the teacher. The teacher will most likely check this progress about once a week.

Use case 10: As a teacher, I can provide general feedback to every student who gets a certain quest wrong. When a teacher notices that many students get a certain question wrong, they can add feedback to it. This feedback will then be visible to all students who have failed the question. This is low priority as it is not needed for the core functionality of the platform, and also low complexity as it should not be too difficult to implement.

Use case 11: As a student, I can make my character. When the student plays the game for the first time they choose what their character looks like. This is low priority and low complexity.

Use case 12: As a student, I can customize my character. Customizability is important for the students to enjoy themselves. This an simple element that brings a lot of added value for the students, making it medium priority. The student should be able to customize their character multiple times per day if they wish to do so.

Use case 13: As a student, I can roam around in the game world. The student should be able to walk around in the game world. This is important enjoyment factor to make the game appealing for the students and is thus marked as medium priority. This is medium complexity because the boundaries of the world need to be properly defined. The students should not be able to walk through buildings or outside their screen. Another factor contributing to this complexity is the walk cycle that links character movement to animations. Students should be able to explore each time they play the game.

Use case 14: As a student, I can solve quests by talking to NPCs and see the impact of my subjects on the real world through visual effects happening to the corresponding district. This is the most important use case for the student, as this defines their experience playing the game. It is also high complexity as when the student solves the quest, realistic consequences must become visible in the game world. Students should be able to solve quests multiple times per day.

Use case 15: As a student, I can remake quests. The student needs to be able to re-attempt quests that were completed before. This is practical for revising exams or revisiting material that the student still struggles with. This is medium priority as it is important for the correct functionality of the platform and the experience for the student. The complexity is low: once participants are able to complete the quest once, enabling them to do so again should not be difficult.

Use case 16: As a student, I can check my progress per subject. It is important that the student themselves can keep track of their progress in the different subjects. They should be able to see how many exercises they have made per subject, their success rate, and how long they spent on the subject. There will also be a global overview where they can see their total amount of exercises completed and their global success rate. This is medium priority as it is a motivating factor for the student to see where they are at in the learning material. It is low complexity to implement, and a student needs to be able to check this multiple times a day if they want to.

As the IT-coordinator, I can create an instance for the school. The IT-coordinator needs to be able to create an instance for the school that is linked to all the students and teachers. This is high priority as all teacher and student features of the platform will happen in this school instance.

Commented [SS85]: Again high?

Commented [FD86]: @Antje Schrurs Ik vind deze zin nogal onduidelijk. Met de zinnen erna denk ik dat het gewoon over collisions gaat? Maar "because the boundaries of the world" leest voor mij als een incomplete zin, bedoel je gewoon: "… because the boundaries of the world need to be properly defined."? En is de walk-cycle (animaties + toetsen die effectief karakter doen bewegen) niet ook een groot deel van de complexiteit?

Commented [AS86R2]: Zo beter?

Commented [FD86R3]: Ja zo vind ik het veel duidelijker, dankjewel.

Commented [SS87]: Is this actually some separate functionality we implement or is this just through the effects?

Commented [AS87R2]: Het kan een aparte functionaliteit zijn (aangezien we deze info ook al tonen aan de teachers, dus tis nie echt moeilijk om de studenten dan ook elk hun eigen progess te tonen)

Use case 18: As the IT-coordinator, I can create teacher and student accounts for my school. These accounts need to be linked to the school instance. The log-ins need to be provided to the teachers and students. This is high priority as everything the teacher and student are able to do on the platform will happen through these accounts. It is low complexity as this has been done by many other platforms before.

Use case 19: As the IT-coordinator, I can configure the class groups at the beginning of the school year. The accounts of the students need to be linked to their respective class groups. These class groups are linked to the school instance. With the existence of the school instance, we can make sure that the same class does not need to be defined multiple times for different teachers. This is high priority as it is important for students to be linked to the correct classes for the operation of the platform.

Use case 20: As the IT-coordinator, I can assign subjects and teachers to classes. The IT coordinator needs to correctly link teachers to the subjects they teach and link the correct subject-teacher combination to each class. This is necessary in order to make everything operate correctly and so it has a high priority.

Use case 21: As the IT-coordinator, I can edit the class configurations. When a student changes class groups, the class that their account is linked to needs to be corrected. This should not occur often, but it does occur in schools every now and then. This is low complexity but an important aspect to realize.

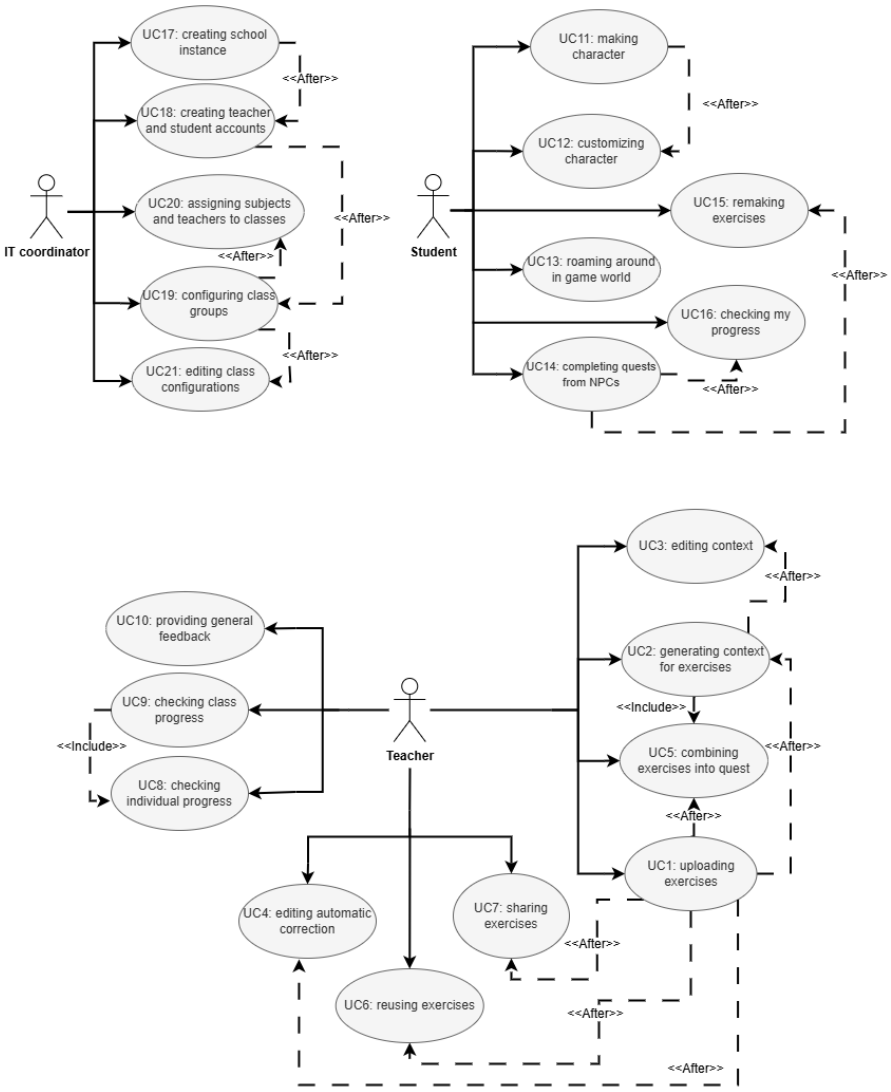
The complexity of realizing all use cases related to the IT coordinator is low as these use cases boil down to the storing of administrative information.

The use case diagram shows all the use cases related to their actor and clarifies the ‘after’ and ‘include’ relationships. In order to keep the diagram readable, more trivial ‘after’ relations are not shown. All teacher and student use cases cannot happen until the IT-coordinator has created the necessary accounts, but this is very obvious and showing these relations would add unneeded complexity to the diagram.

Commented [SS88]: Zou er nog bij zetten dat “The more trivial ‘after’ relationships are not shown on the diagram in order to keep it organized. For example, all teacher and student use cases cannot happen until the IT-coordinator has created the needed accounts.”

Commented [FD88R2]: Zo?

Use case diagram:



Commented [AS89]: Was het oke dat de relaties 'after' noemden? Of moest het iets anders zijn? (vage herinnering van gisteren maar ben niet 100% zeker of het oke was of niet)

Analyzing the list of use cases outlined above leads to the identification of the following three use cases as the most important for realizing this product:

- Use case 1: As a teacher, I can upload a pdf from which the system extracts exercises pairs.
- Use case 2: As a teacher**,** I can generate real-life context for my exercises.
- Use case 14: As a student, I can solve quests by talking to NPCs and see the impact of my subjects on the real world through visual effects happening to the corresponding district.

These use cases manage to capture the essentialities of the platform for the teacher and student sides. Teachers need to be able to upload their exercises and generate contexts for them. These exercises are then presented to the students in the form of quests. By solving quests, the students should be able to link their school material to the real world. This link is partly achieved through the profession of the NPCs involved in the quest and partly through the results of the quest in the game world.

5.2 Quality Attributes

Quality attributes describe the characteristics that determine how well a system performs its intended functions. They focus on how the system should behave and be experienced by its users. Based on the use cases presented in the previous section, a set of key quality attributes was identified as most relevant. The following sections discuss two system-wide quality attributes and one for each of the prioritized use cases.

5.2.1 System-Wide Quality Attributes

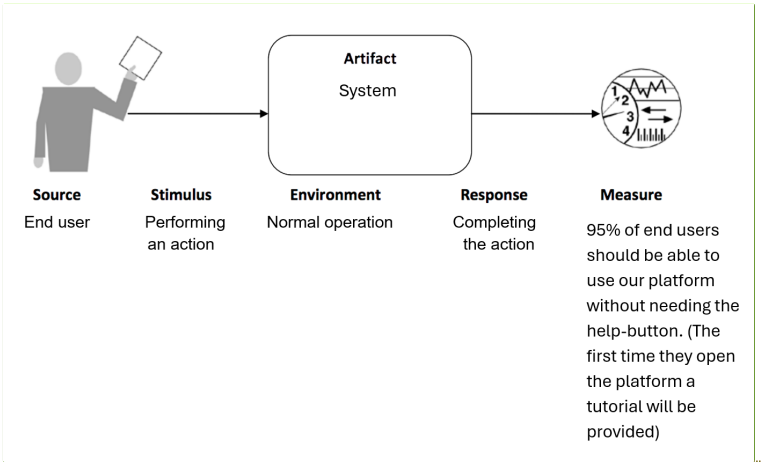
Commented [FD90]: "The two most important quality attributes on system-level and the two accompanying quality attribute scenarios, together with a motivation of the choices made."

Commented [SS91]: Suggestie: "Usability and enjoyability were identified as two key quality attributes on system-level. This section elaborates on these two quality attributes and their accompanying quality attribute scenario's."

Analyzing the list of use cases outlined above lead to the identification of the following two quality attributes as the most important for the system as a whole:

QA1: Usability

The first quality attribute that is important for the system as a whole is usability. The platform should be intuitive, enabling users to work with it regularly with minimal difficulty. The first time the platform is opened a tutorial will show the user how the most important features work. After that the user should be able to work with the platform. In order to measure whether this has truly been accomplished, a help button will be provided through which the user can contact the developers. The platform is usable enough when only 5% of all users need to use this button.



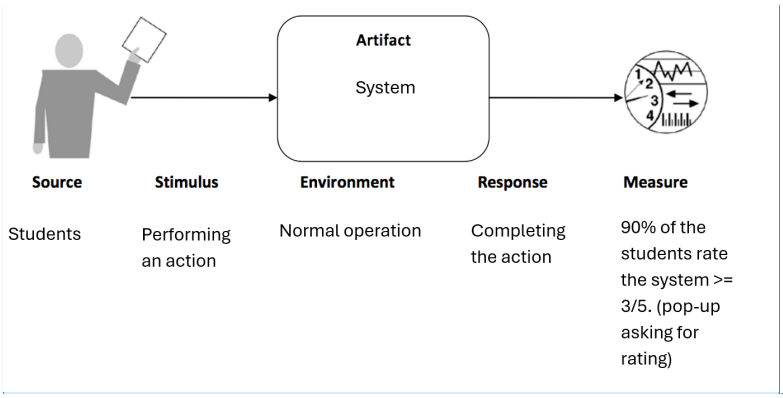
Commented [AS92]: Referencen we deze als figuren?

Commented [SS92R2]: Jup

QA2: Enjoyability

The other system wide quality attribute is enjoyability, which is important mostly for the students side. They need to feel satisfied and enjoy themselves, because this will motivate them to keep on learning. In order to measure the enjoyability of system, the GUESS-18 scale will be used. More information on this scale and other scales that were considered can be found in section 3.6 of the SOTA chapter.

Commented [FD93]: Actually reference SOTA.



Commented [FD94]: Figure has no caption + is not referenced in the text.

5.2.2 Elaboration on most important use cases

Elaboration on UC2

Use Case ID:	2		
Use Case Name:	Context generation		
Date Created:	17/11/2025	Date Last Updated:	24/11/2025

Description:	As a teacher, I can generate real-life context for my exercises.
Actors:	Teacher
Trigger:	Adding a new quest
Preconditions:	1. School instance exists 2. Teacher has an account 3. Teacher is logged in 4. Teacher has already uploaded a pdf with questions and answers 5. The exercises were extracted from this pdf
Postconditions:	1. Exercises have a context around them connected to an NPC and effect within the game world.
Normal Flow:	1. Teacher requests list of extracted exercises 2. UI is updated to show the list 3. Teacher selects exercises from list 4. Teacher presses button to add new quest with the selected exercises 5. System creates new quest containing the selected exercises 6. System generates context for this quest 7. System notifies teacher that generation is completed
Priority:	High
Frequency of Use:	Once every few weeks
Complexity:	High
Assumptions:	Teacher has uploaded exercises before.

Selected Quality attribute for UC2:

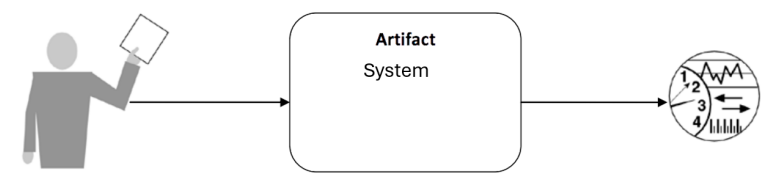
QA3: Accuracy

It is important that the generated context is reliable. Even though this is not an aspect where there is one right answer, it is important to check the quality of the generated content. To be able to quantify this, the teachers judgement will be used as measure. Whenever they are not satisfied with the context and thus have to edit or regenerate it will be counted as not accurate enough. The quality attribute of this use case will be fulfilled if this only happens in maximum 5% of the cases.

Commented [SS95]: Weren't we going to add (re)generation time to the measure?

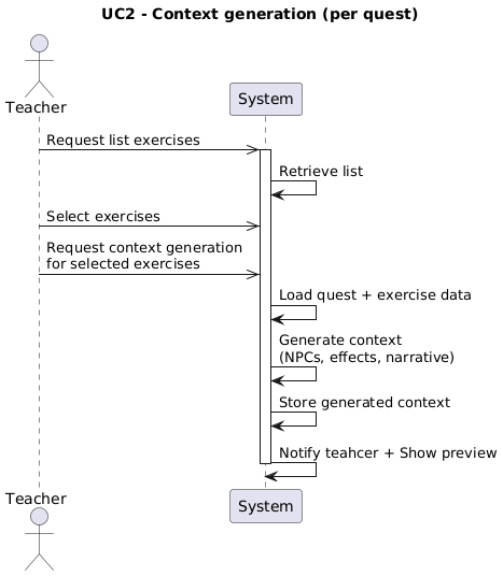
Commented [AS95R2]: Ja klopt, maar hadden we daar een maximumtijd op geplakt?

Commented [SS95R3]: We waren bezig geweest over 45s of 1min maar weet niet meer of dat hierover ging of niet



Source	Stimulus	Environment	Response	Measure
Teacher	Adding exercises	Normal operation	Exercise has context around it	In max 5% of the cases, the teacher should have to edit/regenerate the context.

Commented [FD96]: Figure has no caption + is not referenced in text



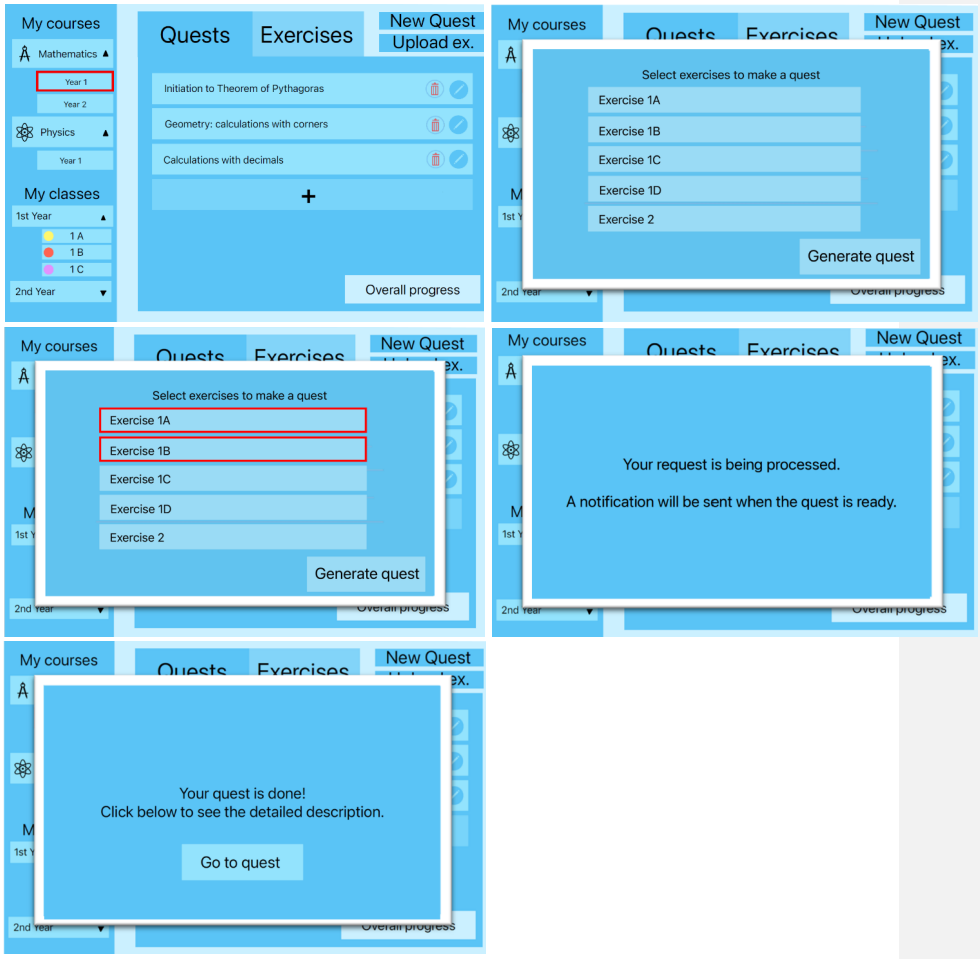
When the teacher wants to create a quest, they first need to request the list with exercises. These are already in the system from when the teacher uploaded them. When the system shows them the list, they can select the exercise(s) where they want to make a quest for. When they then press the 'create quest' button, the quest will be generated.

The mock-up in figure 11 shows this flow.

Commented [AS97]: Overbodig of niet? Ik twijfel omdat het eigenlijk al staat in de template hierboven

Commented [FD97R2]: Zou het laten staan maar zeker het black box diagram referencen

Commented [FD98]: Zou persoonlijk zeggen dat het black-box diagram de flow weergeeft en de mock-up de schermen die de leerkracht zal zien



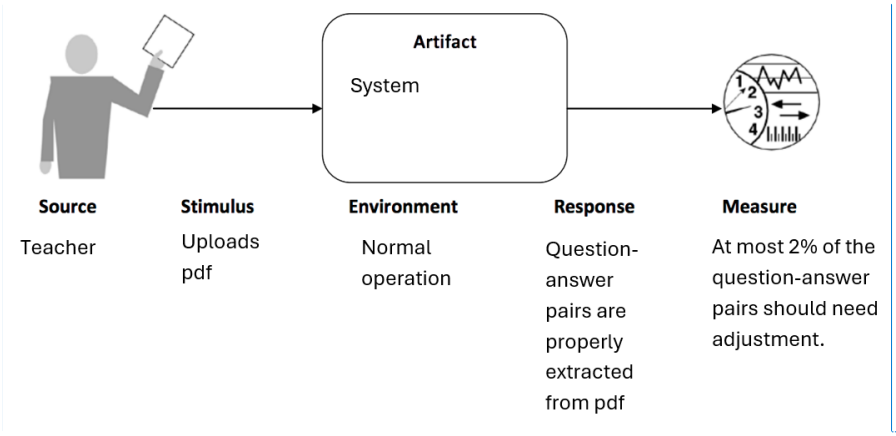
One teacher was contacted to give feedback on these mockups. She found them intuitive enough and said that it is clear what the buttons do. In her opinion, the layout was fine.

Elaboration on UC1

Use Case ID:	1		
Use Case Name:	Extraction of exercises from pdf		
Date Created:	17/11/2025	Date Last Updated:	27/11/2025

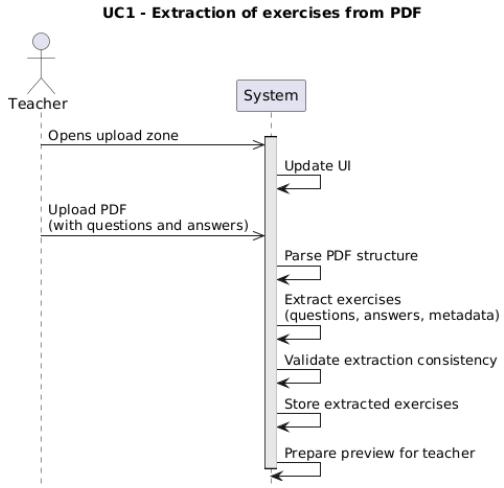
Description:	As a teacher, I can upload a pdf from which the system extracts exercises.
Actors:	Teacher
Trigger:	Teacher uploads a new pdf.
Preconditions:	1. School instance exists 2. Teacher has an account 3. Teacher is logged in
Postconditions:	1. Exercises are in the system in the form of question-answer pairs
Normal Flow:	1. Teacher uploads a pdf with questions and answers 2. System extracts questions and answers from pdf 3. System stores exercises
Priority:	High
Frequency of Use:	Once every weeks
Complexity:	High
Assumptions:	Teacher already has many exercises on pdfs, either from the publisher of the books the school uses or from previous work done.

Selected Quality attribute for UC1:
QA4: Accuracy
Accuracy is an important aspect for the extraction of exercises as exercises in our system should exactly match the provided exercises in the pdf document. Moments where the teacher needs to manually correct the extracted exercises should be avoided as much as possible. Less than 2% of the question-answer pairs should have to be adjusted.

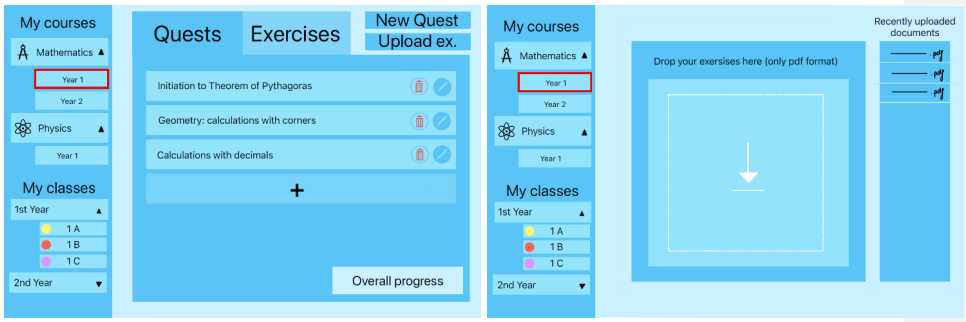


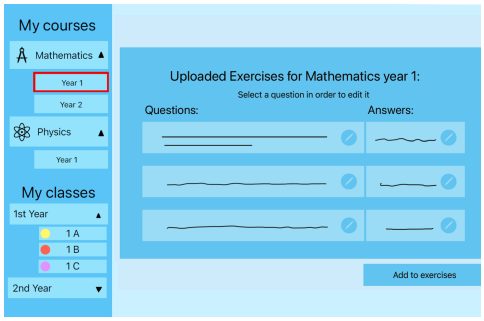
Commented [FD99]: Eigenlijk hebben we niet echt een use case die ons question-answer pairs volledig laat aanpassen. Use-case 4 komt het dichtste in de buurt.

Commented [FD100]: No caption + not referenced in text



Commented [FD101]: Not referenced in text





Elaboration on UC14

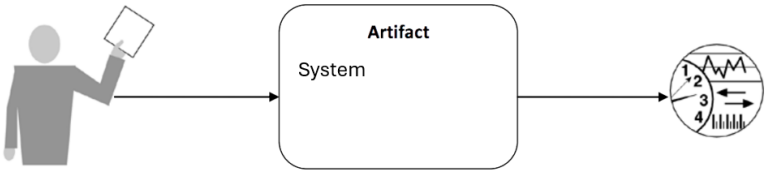
Use Case ID:	14		
Use Case Name:	Quests and their impact		
Created By:		Last Updated By:	
Date Created:	17/11/2025	Date Last Updated:	24/11/2025

Description:	As a **student**, I can solve quests by talking to NPCs and see the impact of my subjects on the real world through visual effects happening to the corresponding district.
Actors:	Student
Trigger:	Selecting a subject's district
Preconditions:	1. School instance exists 2. Student has an account 3. Student is logged in 4. System has a collection of quests for the student 5. Student has navigated to the district of the subject they want to practice
Postconditions:	1. System has student submission registered 2. Student can view their submission
Normal Flow: (correct answer)	1. Student opens quest list 2. System lists quests 3. Student selects quest 4. System shows quest NPC 5. Student fills in their solution 6. Student submits their solution 7. System validates solution 8. System returns V-mark for correct answer 9. System updates visual elements in the game
Alternative Flows: (wrong answer)	1. Student opens quest list 2. System lists quests 3. Student selects quest 4. System shows quest NPC 5. Student fills in their solution 6. Student submits their solution 7. System validates solution 8. System returns X-mark for wrong answer 9. System updates visual elements in the game
Priority:	High
Frequency of Use:	Multiple times a day
Complexity:	High

Selected Quality attribute scenario for UC14:

QA5: Performance

For the quality of the experience for the student playing the game, it is important that the performance is acceptable. Students should never be waiting more than half a second for the effect of a quest to appear.



Source	Stimulus	Environment	Response	Measure
Student	Answers a question of the NPC	Normal operation	Effects in response to the answer	It takes at most 0.5 second between answering the question and the effects taking place

Mock-up

Commented [FD102]: No caption + not referenced in text

Commented [AS103]: @Briek Rombaut please add the mock up here

6 DESIGN PHASE

The Design Phase focuses on translating the requirements identified in the previous phase into a structured software architecture. The goal is to decompose the system into components and define how they interact, using the Attribute-Driven Design (ADD) methodology alongside patterns and tactics to address key quality attributes. This phase includes iterative decomposition at multiple levels using component-connector diagrams.

6.1 Attribute-Driven Design

The Attribute-Driven Design (ADD) methodology is used to systematically transform the system's functional and non-functional requirements into a concrete architectural structure. ADD focuses on using quality attributes, architectural tactics, and design patterns to guide the decomposition of the system into components and to define how these components interact.

ADD proceeds through multiple levels of decomposition. At each level, decisions are guided by the need to satisfy specific quality attributes. Components, responsibilities, and interactions are captured in component-connector diagrams. The architecture becomes more refined as decomposition progresses through ADD0, ADD1, ADD2, and ADD3.

6.1.1 ADD0 and ADD1

The first quality attribute used to divide the system is enjoyability for students (QA2). The portion of the system intended to be enjoyable is the game. Therefore, the game elements are separated from the rest of the system. Within this game component, intuitiveness for students is also emphasized, aligning with the system-wide usability quality attribute (QA1). This focus will be realized in the upcoming ADD diagrams. The game component contains all game logic and the entire game User Interface (UI).

The remaining part of the system is further divided to support usability (QA1) for teachers and IT-coordinators. Achieving this goal requires a user-friendly User Interface (UI). As a first step toward such a UI, the UI is isolated, making later decomposition easier in relation to the targeted usability. Usability will be assessed by checking how many users need to make use of a help button. Ideally, the UI should be intuitive enough that after the tutorial, no more help is needed to use the platform.

The obtained ADD0 and ADD1 diagrams are shown in **Error! Reference source not found.**:

Commented [SS104]: General comment: I took out most informalities (we, our, us). I think we will have to refer to the diagrams more in the explanations. Not all sections describe the responsibilities of the components yet.

Commented [FD105]: Ofwel 2x "component-connector" gebruiken ofwel 2x "component-and-connector". Ik weet wel echt niet welke beter is, iemand een mening hierover?

Commented [SS105R2]: De begeleiders gebruiken component-connector dus ik denk dat.

Commented [SS106]: Is this true?

Commented [FD106R2]: Ja want das via MVC

Commented [SS106R3]: Ja maar dat doen we niet voor de game en de zin ervoor spreekt over intuitiveness in de game

Commented [FD106R4]: Mijn excuses, heb verkeerd geantwoord op je comment. Is wel zo dat we in Game zeggen dat we QA1 usability realiseren met ECS, dus zou deze tekst laten staan.

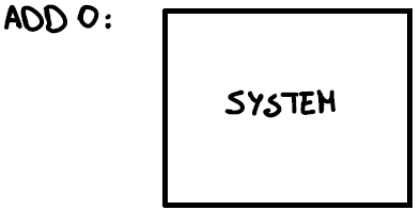
Commented [SS106R5]: Ahh juist ja ECS is ook voor usability goed

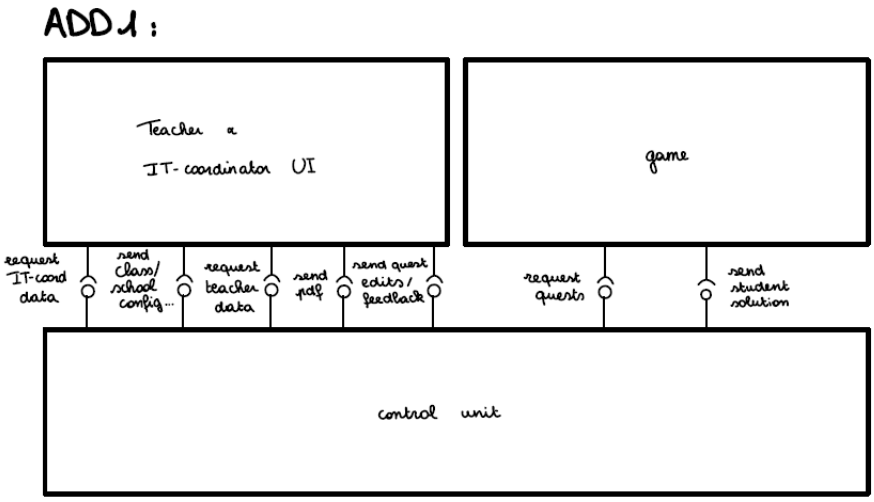
Commented [SS106R6]: Dan is het in orde!

Commented [FD107]: @Saira Dit ok?

Commented [SS107R2]: Jup, ziet er goed uit!

Commented [FD108]: Fix reference





In this decomposition, the teacher & IT-coordinator UI is responsible for all direct interactions with these types of users. The game is responsible for all game logic and game interactions. That means that all interactions with the students are via the game. The control unit is where the unique functionality of the platform is situated. In the control unit, exercises are extracted from pdfs, context is generated for exercises, and all data (except game visuals) are stored inside the system core.

6.1.2 ADD2

The components identified in ADD1, namely the Teacher & IT-coordinator UI, the game, and the control unit, each contain substantial internal complexity and exhibit distinct quality attribute concerns. For this reason, each component requires further decomposition in ADD2. The following subsections describe the ADD2 decomposition for the Teacher & IT-coordinator UI, the game, and the control unit.

6.1.2.1 Teacher and IT-coordinator UI

Commented [SS109]: Description of responsibilities needed? Or at least explanation of the diagram?

Commented [FD109R2]: Goed punt, beter nu?

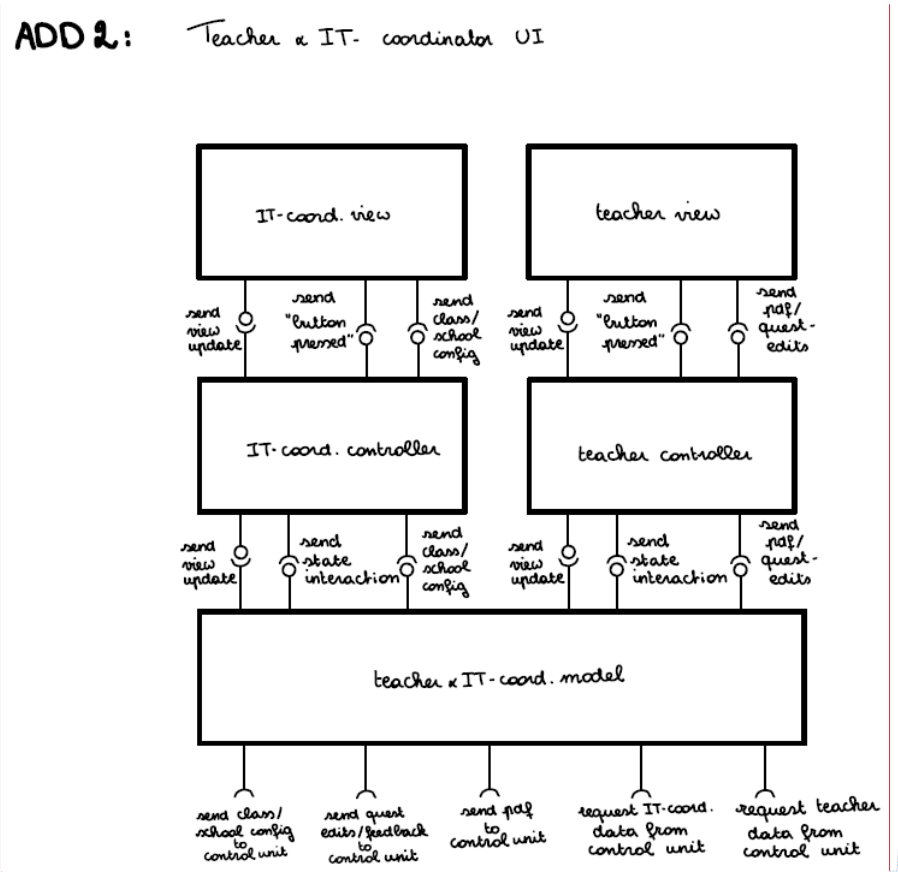
For the Teacher and IT-coordinator UI, the Model-View-Controller (MVC) pattern is used. MVC is a software design pattern that effectively supports the realization of usability (QA1), as it separates the

view from the underlying data logic. This separation enables the development of an interactive application with a flexible human–computer interface that is designed to be as easy to use as possible. The flexibility given by the MVC pattern also allows to more easily change the user's view during future testing, which will allow to more easily work towards the most user-friendly application possible.

The obtained ADD2 diagram is shown in **Error! Reference source not found.**. In this diagram, the model component is responsible for retrieving all necessary data. Data that is relevant for both the game and the teacher & IT-coordinator UI is stored in the control unit. When this data needs to be viewed by a teacher or IT-coordinator, they interact with buttons within their respective views. These buttons call functions within the relevant controller. This controller then asks the model for the necessary data. The model interacts with the control unit to retrieve this data, and sends it back towards the controllers who then change the views. There is only one model to insure that there is only one component that has to interact directly with the control unit.

Commented [FD110]: Fairly certain that figure should become figure 15 or something.

Because usability for the teacher and IT-coordinator take on different forms and because teachers and IT-coordinators do not share views, separate views are created for each of them. This separation allows the design to focus on the specific usability needs of each user persona. These views display information to the user and contain all necessary interactive components such as buttons. When these interactive components are activated, they call upon functions within the relevant controller. Because the functionality for teachers and IT-coordinators is very distinct and because they have a different view they have their own controllers. These controllers are responsible for handling user input.



6.1.2.2 Game

To satisfy the assigned quality attributes QA1 (usability), QA2 (enjoyability), and QA5 (Performance), a decision was made to focus on performance and separation of concerns. At its core, it is designed to employ Data-Oriented Design (DOD) as its primary strategy through the use of the Entity-Component-System (ECS) pattern commonly known for its strong reusability, decoupling capacity, and performance. This decision can be motivated by looking at industry giants: Unity's DOTS and UE5's MassEntity, both of which have adopted the ECS pattern for high-performance use cases. One of the reasons ECS is preferred for games is its cache efficiency, as games most often consist of a large number of data components needing frequent access. The ECS pattern is facilitated through an event system to further improve the separation of concerns and reduce the potential call stack depth.

The obtained ADD2 component connector diagram is shown in **Error! Reference source not found.**. The event system forwards events to other components who have registered themselves as listeners

Commented [SS111]: Is this diagram now correct?

Commented [FD111R2]: Looks good to me

Commented [SS112]: Explanation needed?

Commented [FD113]: @Adrien Van Lancker Heb dit lichtjes aangepast gebaseerd op de feedback op ons document. Vind je dit ok?

Commented [FD113R2]: Heb ook de tekst hierna verder uitgeschreven gebaseerd op ADD2-GAME.pdf, feedback daarop is ook zeker welkom

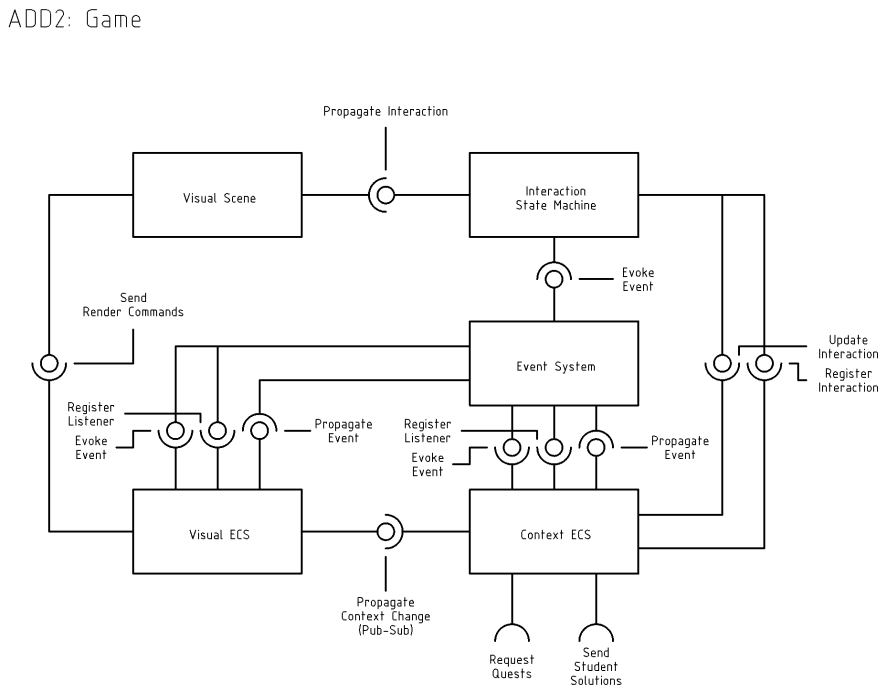
Commented [AL113R3]: Was inhoudelijk allemaal juist, heb enkel maar stijl aanpassing uitgevoerd

for these events. All events with a direct effect on game logic are forwarded to the context ECS and all events with a direct effect on the rendering are forwarded to the visual ECS. The event system reduces call stack depth normally caused by entities recursively interacting with each other. When interactions are routed through the event system there is a constant call depth, which improves the performance (QA5).

Reactive and decoupled visualization is accomplished by using a separate ECS for the visual logic which observes relevant changes in the state of the context ECS through a publish-subscribe pattern. This reduces the required knowledge of the game logic for frontend developer and makes it easier to develop individual visual components that can be replaced by alternatives to test their performance on visual usability and enjoyability benchmarks. The visual ECS evokes and listens to events to drive visual logic but mainly reacts to changes in the context ECS via its direct interconnection and holds the visual reactive state. The context ECS holds the world's state and logic. It also builds the Interaction State Machine graph at runtime by registering and updating interactions.

A dedicated Interaction State Machine describes and captures all possible interactions that the player can make and invokes representative events. Any interaction that the player can make along with its metadata (such as whether the interaction is available and which event(s) it evokes) is registered here.

The visual scene is the traditional scene-graph/render-pipeline. It contains components such as game objects, meshes, lights, and cameras, driven directly by the visual ECS. The visual ECS writes into this scene with a one-way data flow. The visual scene is purely for representation of the game.

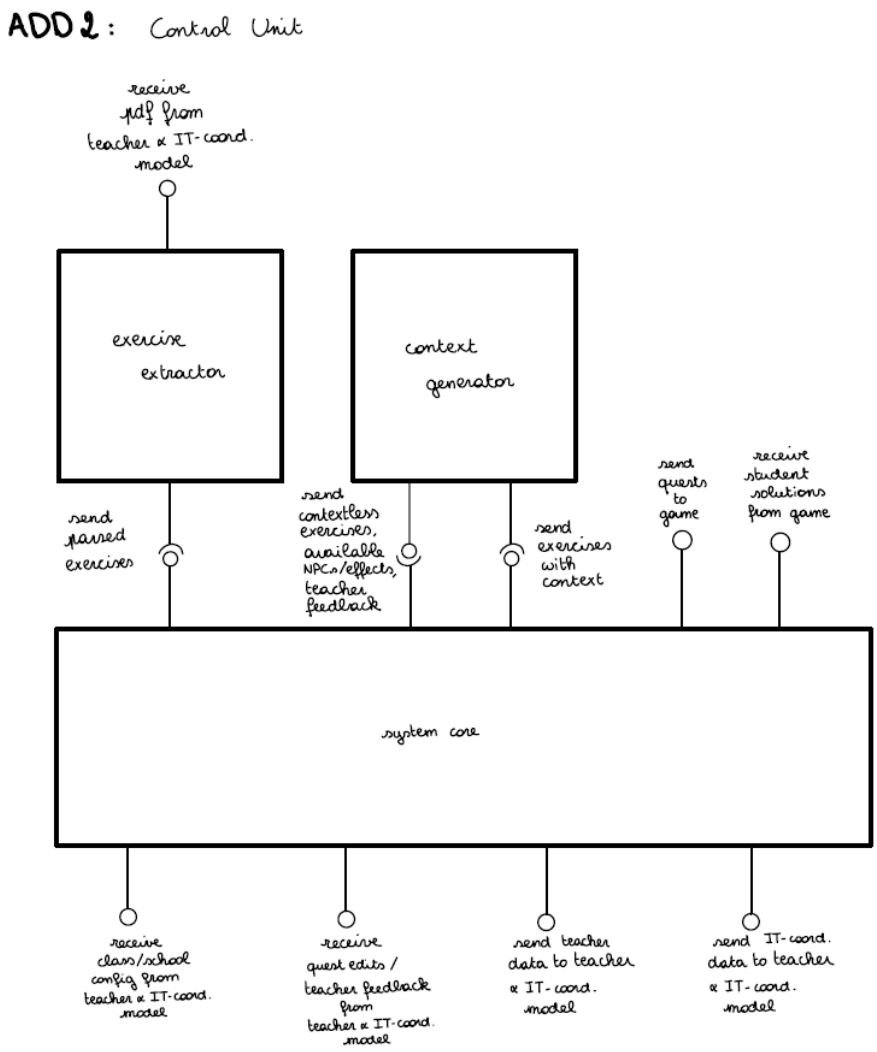


Commented [FD114]: Figure should have higher number + should be referenced in text

6.1.2.3 Control Unit

For the control unit, concerns are separated to allow focused attention on the different aspects of the quality attributes. To satisfy the third and fourth accuracy-related quality attributes, the control unit is further divided. This is shown in **Error! Reference source not found.**. Accuracy for the PDF extractor and accuracy for the context generator represent fundamentally different concepts, making it important to isolate these components from the rest of the control unit. This separation enables targeted attention to the specific accuracy requirements of each component.

Commented [FD115]: Figure should have higher number



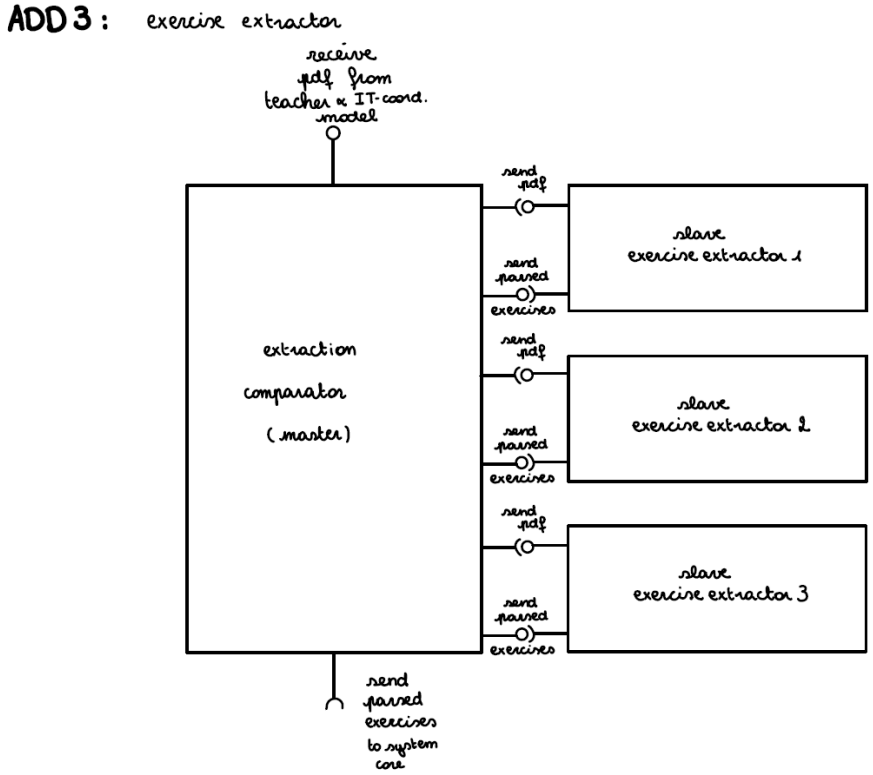
6.1.3 ADD3

The ADD2 decomposition identifies internal subsystems within the Teacher & IT-coordinator UI, the game, and the control unit. Some of these subsystems contain critical quality attribute challenges that require an additional level of refinement. ADD3 focuses on these areas, using more specialized design tactics and patterns when necessary. At this level, decomposition is driven by attribute-specific concerns, for example, ensuring extraction accuracy in the exercise extractor or controlling the quality and reliability of generated context in the context generator. ADD3 therefore introduces detailed

internal structures, refined workflows, and selected architectural patterns to meet these specific quality attributes. The subsections below describe the ADD3 decomposition of two components within the control unit: the exercise extractor and the context generator.

6.1.3.1 Exercise Extractor

To satisfy the accuracy quality attribute related to extracting exercises from PDFs, the exercise extractor component is further divided, using an ensemble of extractors as shown in **Error! Reference source not found.**. The master–slave design pattern is applied: the master component distributes PDFs to the slave extractors, which return the extracted questions and answers. The master then compares the outputs and applies majority voting to determine which parsed exercises are passed on to the system core component.



6.1.3.2 Context Generator

The context generator is further split up in order to satisfy the accuracy quality attribute (QA3). It is decomposed according to the supervisor pattern as can be seen in **Error! Reference source not**

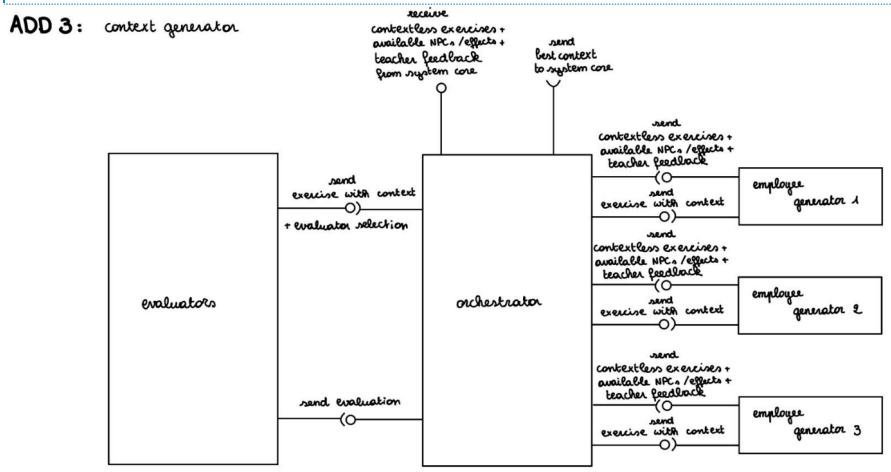
found.. An orchestrator will orchestrate two groups of workers. The first group of workers, denoted with the term employee on the diagram below, are agents who receive contextless exercises along with the available NPCs and effects and a persona from the orchestrator. A persona is a set of instructions intended to shift the attention of the agent such that it more strongly focuses on certain knowledge (e.g., “You are a mathematician.”).

Then each employee generates a context around these exercises using the available elements and send their result back to the orchestrator. The orchestrator performs basic sanity checks, such as “Does this context only use NPCs from the provided list?”, “Does this context contain effects only from the provided list?” Any context that fails these basic checks is dropped.

The orchestrator selects an ensemble of a second type of worker agents, the evaluators. The contexts that did not fail the basic checks are passed along to these evaluators, who return an evaluation of the context. When all evaluation agents have provided their scores, they are aggregated to produce a final context ranking. The best context is returned to the system core, where it is stored.

When a teacher want to regenerate a context, they will be able to check boxes or leave a comment that explains why the given context was not good enough. This feedback will be sent along with the exercises to the orchestrators, who passes along the feedback to the employees. By including this feedback, the regenerated context has a higher probability of reflecting the needs of the specific teacher.

Commented [FD116]: Should be changed to reflect latest idea (only best context goes to system core)



6.2 White-box Sequence Diagrams

In this section the white-box sequence diagrams for the top three use cases are given and explained. The full explanation of these use cases and their black-box sequence diagrams can be found in section 5.2.2.

6.2.1 White-box Sequence Diagram UC2

This white-box sequence diagram gives a detailed overview of the workflow of use case 2. This use case is about generating real-life context for one or more exercises which then together form a quest.

The teacher starts by pressing a button to indicate that they want to create a new quest. This starts an event to retrieve the list of exercises from the system core. When this list has been retrieved, the teacher's view is updated to visualize it. Then the teacher selects the exercises that should be in the quest from the list of exercises that are within the system. As the teacher selects exercises from the list, their view is updated so that they can see which exercises they've selected.
When the teacher is done selecting exercises, they press a 'create quest' button. This immediately updates their view to show that context generation is in progress and inform the user that they will be notified when the generation is done. Meanwhile, the IDs of the selected exercises are send towards the system core, who retrieves these specific exercises and sends them along to the orchestrator within the context generation component.

The orchestrator selects personas which match the exercises within the quest. These personas and the exercises are sent to the employees who then attempt to generate a context. When these contexts are returned to the orchestrator, the orchestrator first performs some basic sanity checks and drops any context that fails these. If the dropping of obviously faulty contexts causes the total amount of contexts to be too small, the workers are given the same task again.

The orchestrator selects its ensemble of the evaluators based on the content of the exercises. The contexts are passed along to these evaluators, who return an evaluation of the context. When all evaluation agents have provided their scores, they are aggregated to produce a final context ranking. The best ranked context is returned to the system core, where it is stored.

When the quest has been saved in the system core, the system core notifies the teacher & IT-coordinator UI. This component then updates the teacher view so that the teacher can see the notification on their screen and knows that generation is complete.

Actually looking at the generated context is part of a different use case, namely use case 3. At the end of this workflow the teacher does not see the context but only gets notified that it has been made. Clicking on this notification will of course send the teacher directly to the preview of the quest, where edits can be made.

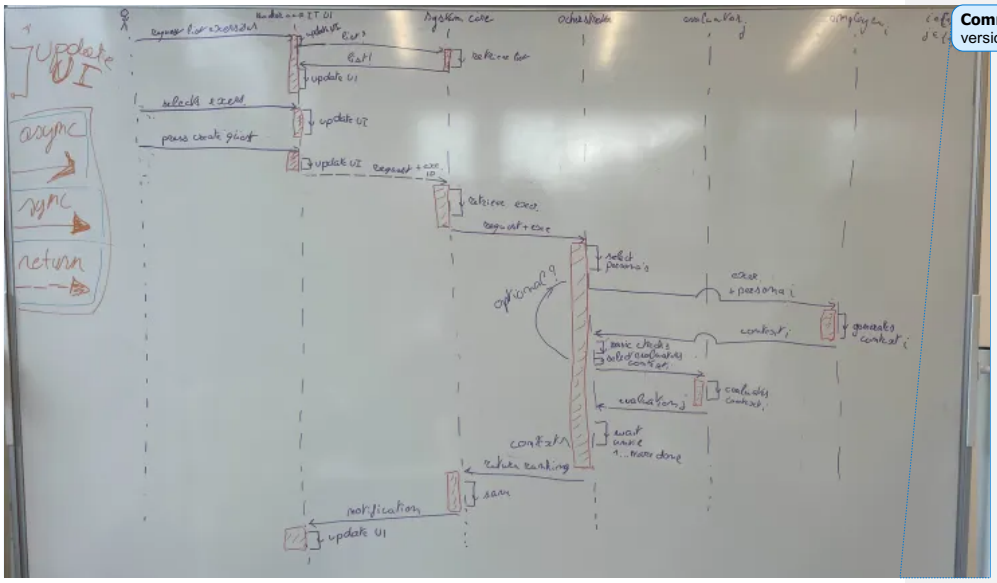
Commented [FD117]: Ik moet in deze zin nog echte reference naar figuur zetten.

Commented [SS118]: Ik zou zeggen the workflow of the flow, the working klinkt wat verengelst vanuit nederlands

Commented [FD119]: Denk dat dit nu als "optional?" in ons schema staat maar denk dat we hierop de commentaar zullen krijgen "wat als je keer op keer opnieuw begint ? + wat is maximale uitvoertijd ?". Dus mss moeten we dit nog wat verder uitdenken?

Commented [FD119R2]: Zou dit (en dus ook de 'optional?') mss gewoon weglaten ? Maar dan gaan ze mss vragen "what if all contexts fail the sanity checks and are dropped?"

Commented [SS119R3]: Ja dat kan wel. Als ze dat vragen, kunnen we zeggen dat de teacher een notification krijgt dat de generation is gefaald en dat die moet opnieuw proberen. Of we laten de optional en als je meer dan drie keer opnieuw moet proberen, geef je die melding van het is niet gelukt.



Commented [FD120]: Should be replaced by the neat version

6.2.2 White-box Sequence Diagram UC1

This white-box sequence diagram gives a detailed overview of the workflow of use case 1. This use case is about uploading a pdf from which exercises are extracted.

Commented [FD121]: Ik moet in deze zin nog echte reference naar figuur zetten.

Commented [SS122]: Idem, zou workflow of flow gebruiken

The teacher starts by pressing a button to go to the upload-zone. Pressing this button causes the UI to update so that the teacher gets to see the upload-zone where they can upload their pdf document. This upload-zone is pictured in the mockup in section 5.2.2. The uploaded pdf is sent from the teacher and IT-coordinator UI towards the extraction comparator component within the exercise extractor.

Commented [SS123]: Ik denk dat dit eigenlijk twee stappen zijn, ga naar uploadzone => ik zie nu upload zone en klik op upload => ik krijg folder structuur. Of niet?

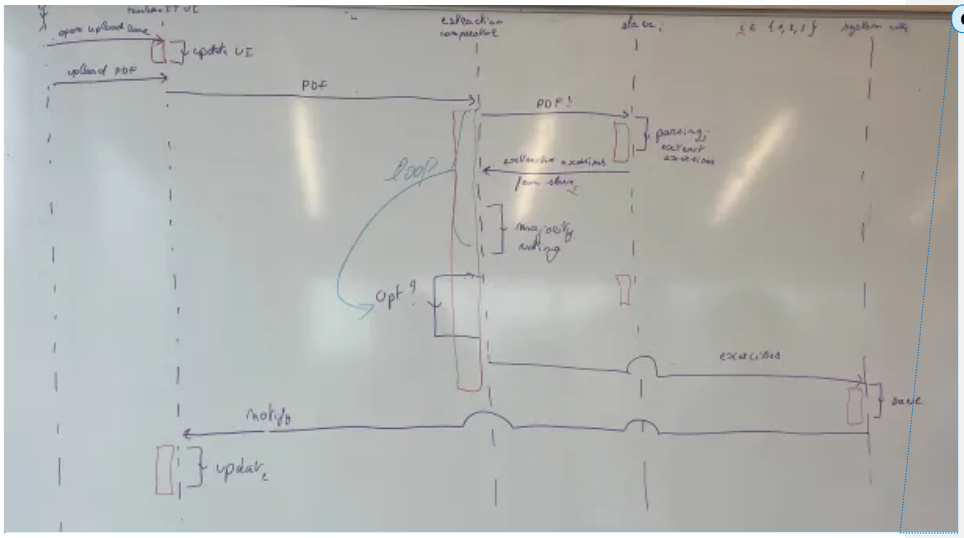
Commented [SS123R2]: Allez in ons diagram is het 1 stap, maar in het echt is het meestal opgesplitst denk ik.

Commented [FD123R3]: Ja eigenlijk moeten we dit mss nog eens bespreken. Ik dacht wel half: is eigenlijk makkelijker om hier uit te leggen als ze gewoon op een knop drukken en direct die folder popup zien, want er is toch geen echte 'extra informatie' die ze moeten krijgen in de uploadzone

The extraction comparator forwards the pdf to each of its slaves. These slaves parse the pdf and extract the exercises. Their results are sent back to the extraction comparator. There are 3 slaves. If 2 or more of these slaves have the same result for the parsing of an exercise, then this exercise is saved. If all slaves got a different result then they all try again to parse this specific exercise. When all exercises are extracted, they are sent to the system core for storage. When the system core has saved the exercises it notifies the teacher & IT-coordinator UI so that the teacher receives a notification that the exercises have been extracted from the pdf. The UI update at the end is simply showcasing this notification.

Commented [FD124]: Denk niet dat we dit besproken hebben voor individuele exercises, maar het zou wel zot zijn als we heel de pdf opnieuw parsen voor een klein foutje, toch?

Commented [SS124R2]: Goed punt, denk dat indd heel de pdf opnieuw parsen overdreven is



Commented [FD125]: replace

6.2.3 White-box Sequence Diagram UC14

This white-box sequence diagram gives a detailed overview of the workflow of use case 14. This use case is about students being able to complete quests.

Commented [FD126]: Ik moet in deze zin nog echte reference naar figuur zetten.

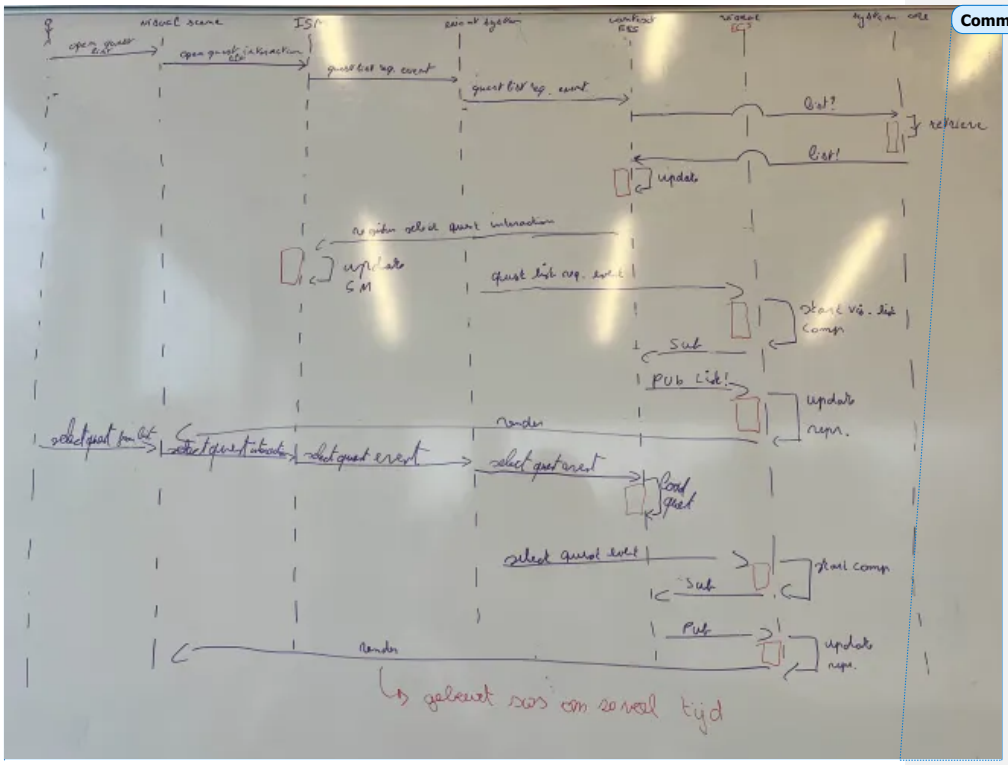
Commented [SS127]: idem

To start a quest, students must first open their quest list. When they press the ‘quest list’ button in the visual scene, the visual scene sends the Interaction State Machine (ISM) a message that contains the information of which keys were pressed (left mouse button) and where the cursor position was at that time. The ISM interprets this information and recognizes it as the ‘open quest list interaction’. Because this interaction is available when the student sees the ‘quest list’ button, the ISM registers an ‘open quest list event’ in the event system.

The event system then notifies all listeners of ‘open quest list events’, which are the context ECS and the visual ECS. There is space left in between the ‘open quest list event’ being sent to the listeners on the diagram, but that is purely to make the diagram easier to follow. The two ‘open quest list event’ arrows starting from event system can happen in any order, and their consequences activated by the context ECS and the visual ECS can happen in parallel.
When the context ECS is notified about this event, it retrieves the list of quests from the system core. When it has received this list, it temporarily saves a copy so that the chosen quest does not need to be retrieved from the system core again. It then informs the ISM that the interaction ‘choose a quest’ is now available and a quest can be selected from the list. The ISM updates itself to reflect the availability of this interaction.
Meanwhile, when the visual ECS is notified about the ‘open quest list event’, it starts a new visual component to represent the quest list. This new component subscribes to the context ECS. At some point after the context ECS has saved a copy of the quest list, it publishes the necessary information from the quest list (e.g.: quest titles / chapters covered in the quest) to this new visual component within the visual ECS. This visual component updates itself so that it contains the correct representation of the quest list.
When the next ‘render’ message is sent from the visual ECS to the visual scene, the player sees the quest list open. The ‘render’ message is always sent from the visual ECS to the visual scene when a timer runs out, so that the visual scene updates at regular intervals.

At this point the player can see the quest list. In a next step, the player clicks on a quest item from this list to start a quest. What happens next is very similar to what happened when the player opened the quest list. The visual scene sends the ISM a message that contains the information of which keys were pressed (left mouse button) and where the cursor position was at that time. The ISM interprets this information and recognizes it as the ‘select quest interaction’. The ISM registers the ‘select quest event’ in the event system, which notifies both the context ECS and the visual ECS.
The context ECS loads all necessary metadata from the selected quest and publishes this for a visual component. The visual ECS starts a new visual component and subscribes this component to the context ECS. When this visual component receives the necessary data (NPCs, text that should be displayed, locations of text bubbles, …) it updates itself to prepare all new visuals. Afterwards, when the next ‘render’ message is sent out, the visual scene makes the quest visible.
Now the student is inside the game loop for this specific quest and can really start playing the game. The game loop is illustrated in a separate diagram for clarity.

Commented [FD128]: Check of dit nog steeds waar is met de nieuwe diagrammen



Commented [FD129]: replace

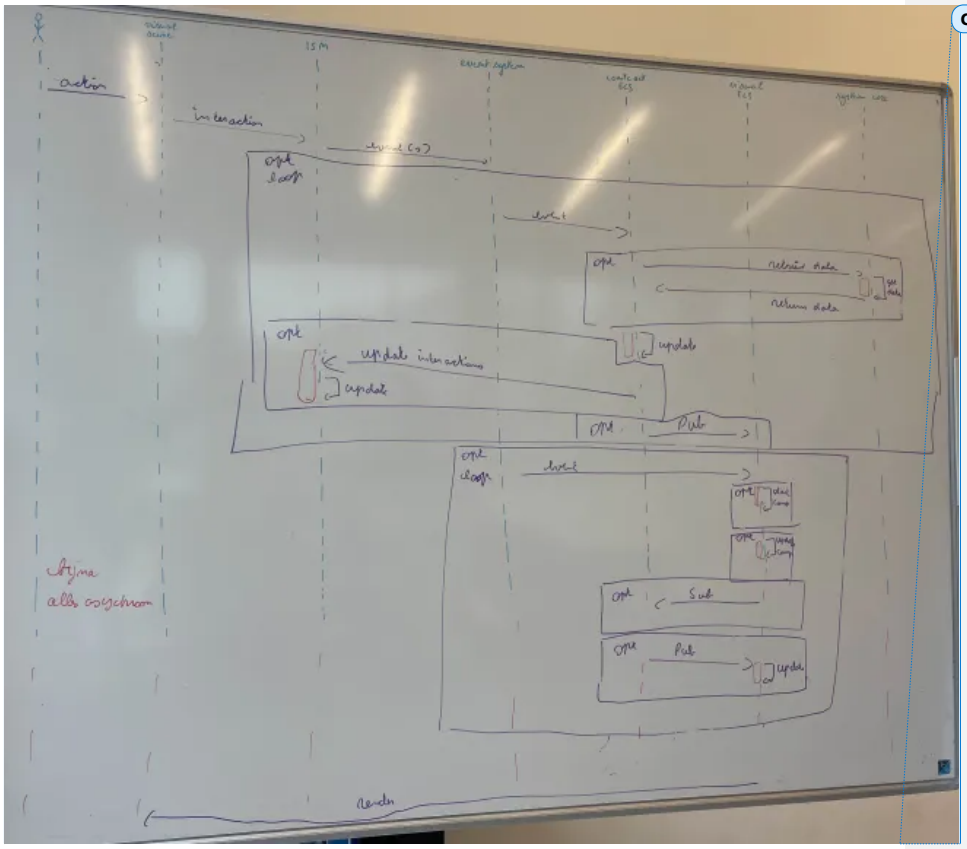
The diagram below showcases the game loop in all generality. The reason that we do not go into specific detail in this workflow is because it depends heavily on the quest. First, the student will have to walk towards the correct NPC to start the quest. Each step that the student takes is a button press that will be registered in the visual scene which will then inform the ISM. The ISM will interpret each button press combined with the current mouse cursor position. If this is a valid interaction, then the ISM will notify the event system that one or more events are taking place.
For each event there are three possible situations:

- Both the context ECS and the visual ECS are subscribed to the event
    - These situations are analogous to those described in the workflow of opening the quest list and selecting a quest. Both the context and its visual representation changes.
- Only the context ECS is subscribed to the event
    - Something about the context changes in a way that does not immediately need to be visualised.
- Only the visual ECS is subscribed to the event
    - This can be the case for very small visual effects, such as weather effects which only start when the player reaches a particular boundary (walks outside). Small effects like falling snowflakes may become visible on the screen without affecting gameplay.

Commented [FD130]: I'd like to add an example here but find it difficult to think of one

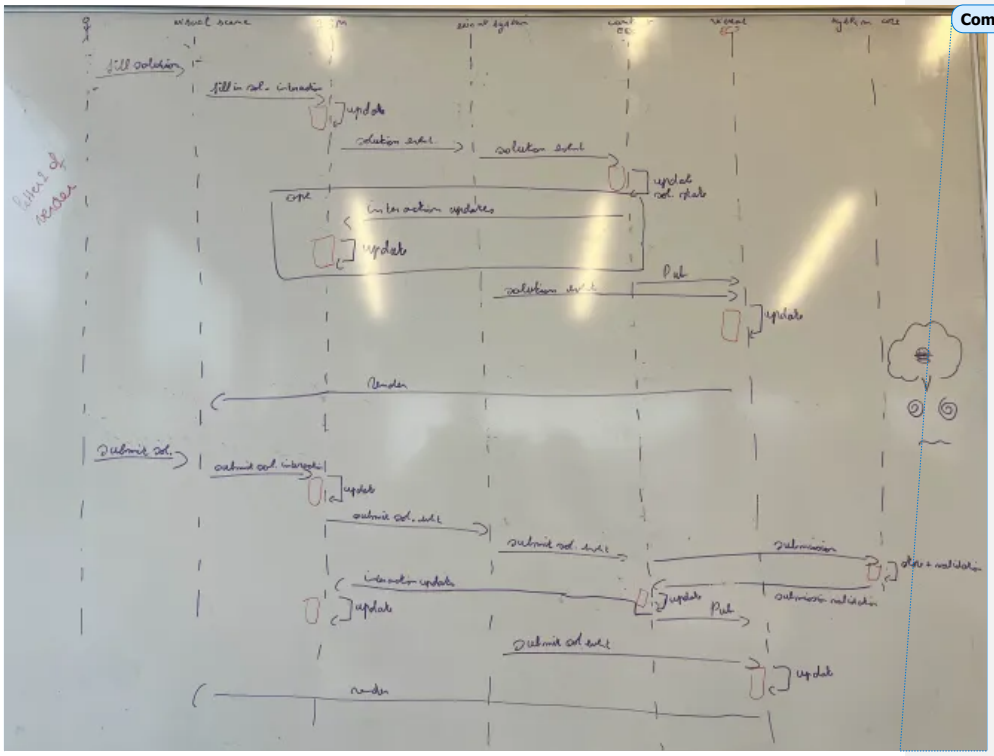
Commented [FD130R2]: @Adrien Van Lancker any ideas?

In any case, either or both the context ECS and the visual ECS react to the event.
When the context ECS reacts to an event, it may need to retrieve additional information from the system core. If that is the case, this happens before the context ECS updates itself in response to the event. It is possible that the update inside the ECS causes new interactions to become available or causes available interactions to become unavailable. If that is the case, the context ECS notifies the ISM about this. It is also possible that there are visual components which are subscribed to data from the context ECS that has now been updated. If that is the case, the updated data is sent to these subscribers.
When the visual ECS reacts to an event, it may start new visual components or update existing ones. It is possible that these new or updated components now need to be subscribed to specific information from the context ECS. Subscribes may receive data from the context ECS and update themselves accordingly.
At some point, the visual ECS sends a new 'render' message to the visual scene. This render message is sent every time that a timer runs out. Only after this render message do visual changes become visible to the player.
By completing actions in the game loop, the player eventually reaches a screen where they need to fill in the solution of an exercise. For this, we have drawn another detailed diagram.



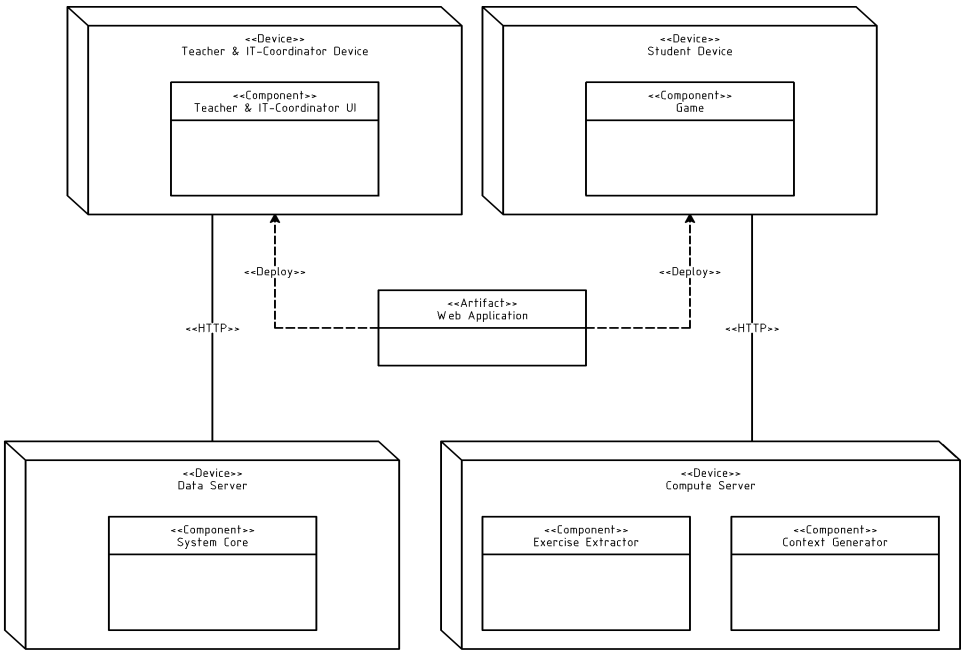
Commented [FD131]: replace

When the player starts filling in their solution, each button press again causes a new interaction to be started. Each new letter will make the ISM send out a new ‘solution updated event’. Both the context ECS and the visual ECS are listeners for this event. The context ECS saves the solution state and publishes this to the visual component that visualizes the typed out text. If the solution state reaches its character limit, the context ECS notifies the ISM that typing in a new letter on the keyboard should no longer cause a ‘solution updated event’. Typing backspace will delete a letter from the solution and make the ‘solution updated event’ available again. This makes sure that there is no overflow in the buffer where the (partial) solution is stored.
The visual ECS updates the visual component that showcases the (partial) solution, so that on the next ‘render’, the student can see all letters that they have typed out so far.
When the student is done with typing the solution, they press a ‘submit’ button. This button press is registered by the visual scene who then notifies the ISM. The ISM makes sure that the student can not type any extra letters by disabling the ‘solution updated interaction’. Then, the ISM sends out a ‘solution submitted event’ to the event system. Both the context ECS and the visual ECS are listeners for this event and will react to it.
The context ECS will send the saved solution to the system core for review. This is done because the context ECS executes on the device of the student and can thus be more easily compromised. For that reason, it is preferable to let the system core check whether a solution is correct.
When the system core has checked the submitted solution against its saved correct solution, it updates its statistics to reflect the student’s progress and replies to the context ECS. This reply contains the information of whether or not the student answer was correct. If the quest was solved correctly, the context ECS moves the quest to the list of ‘fully completed quests’. Otherwise, the quest stays in the student’s quest list, so that it may be reattempted at a later time. The context ECS publishes the information of whether or not the solution was correct to the correct visual component(s) in the visual ECS. These visual components update themselves so that the effects of succeeding or failing at the quest become visible in the game word upon the next ‘render’.



Commented [FD132]: replace

6.3 Deployment diagram



WIP-Adrien: Description.

Commented [AL133]: @Adrien Van Lancker this is something you still have to complete.

7 TECHNOLOGIES

This chapter describes the concrete technology choices made to realize the prototype defined in Chapter 8. While the State-of-the-Art analysis (Chapter 3) explored relevant research directions and candidate technologies, this chapter translates those insights into explicit architectural and implementation decisions.

For each major subsystem, multiple feasible technology options are compared. A final choice is then motivated based on a fixed set of design properties. The goal of this chapter is to make the system understandable, implementable, and estimable for development.

7.1 AI Processing Layer

The AI Processing Layer forms the computational core of the platform, as it is responsible for both extracting structured exercises from uploaded documents and generating narrative contexts that connect these exercises to the game world. Because this layer directly processes copyrighted educational material, its design is strongly constrained by privacy and deployment considerations.

In addition, this layer directly affects teacher experience: long or unpredictable processing times would break the intended workflow of uploading, previewing, and iteratively refining content. As a result, the deployment setting and model type are critical architectural decisions.

Sixteen teachers were asked about their opinions concerning Artificial Intelligence (AI) via a questionnaire. All the teachers were open to the use of AI in an educational context. Twelve teachers already use AI for their work in some way. Thirteen teachers are willing to trust an AI model to extract their exercises. Eleven teachers are willing to let an AI model generate context around their exercises. This confirmed the importance of the AI part in our application.

Commented [F[134]]: Probleem: ik begin hier plots over AI terwijl dit voor de formuleren van ons idee staat. Verzet ik dit naar verder in de deliverable, bv bij de technologies chapter? Want hier komt AI precies uit de lucht gevallen.

Commented [SS134R2]: Ja denk dat dit dan indd beter bij technologies past

Commented [F[135]]: @Frederieke Delchambre Zet waar het het beste past in de technology chapter.



Considered options:

Option	Privacy	Latency Control	Customization	Decision
External AI API	Weak	Low	Low	✗
Private cloud AI[1]	Medium	Medium	Medium	✗
Local open-weight model	Strong	High	Medium	✓

Commented [AL136]: What is this, not really clear.

Decision:
The comparison shows that external AI APIs are fundamentally incompatible with the privacy requirements of the project, as they require uploading educational content to third-party infrastructure. Even private cloud solutions introduce additional operational and legal complexity while still limiting fine-grained control over inference behavior. Local open-weight models, in contrast, allow full control over data flow, inference scheduling, and prompt design. This makes them the only viable option within the constraints of publisher content protection and university-hosted deployment.

Once local inference is selected, a second design decision concerns the scale and family of the language model itself. While recent State-of-the-Art models demonstrate impressive generative capabilities, their suitability for this project depends less on raw fluency and more on controllability, latency, and deployment feasibility.

[1] Private cloud AI: Running AI models on dedicated, isolated cloud infrastructure that an organization fully controls, so sensitive data stays private while still benefiting from cloud-scale compute and storage.

The evaluated model families span a wide range of parameter sizes, from small-to-medium scale models that can run on a single GPU to very large models that require parallelization techniques.

Candidate model families:

Model family	**Parameter scale**	**Strengths**	**Weaknesses**
GLM-4.1V-9B-Thinking	9B	Easy to deploy on a single GPU. Lower memory and compute requirements. Decent at associative tasks. Low latency.	Lower generation quality, especially on more complex and specialized problems.
MiniMax-M2	230B (10B active)	Better generation quality, with lower compute. Lowest cost-to-performance ratio. Low latency.	Higher memory requirements. Complex sharded deployment.
Active large active LLMs	100B+ active (families without MoE)	Best theoretical generation quality. Lower probability of domain knowledge bottlenecks.	Highest cost-to-performance ratio. Highest latency.

For teacher-facing workflows (PDF upload, preview, regeneration), predictable response time is more important than marginal output quality. Larger models with a bounded number of active parameters (≈10B parameters) have faster inference time, while still achieving sufficient quality.

Final decision: A large-scale open-weight model (e.g. MiniMax-M2) will be used for the primary task of model orchestration and context generation, while smaller open-weight models (e.g. GLM-4.1V-9B-Thinking) will be used for context evaluation and exercise extraction. Both models will be self-hosted to meet legal privacy and data management constraints.

7.2 PDF Extraction Technology

Extracting exercises from PDF documents is a notoriously difficult task, especially in STEM domains where layout, mathematical notation, and visual structure carry semantic meaning. PDFs do not preserve reading order, and identical logical structures can be represented very differently across publishers and documents. As a result, singular extraction tools often fail to recover the intended structure of exercises, which requires combining various extraction tools.

Final Decision:
Pure PDF-to-text approaches are attractive due to their speed and simplicity, but no single tool succeeds in extracting information across documents with very different structures and representations. A hybrid approach is thus chosen which uses various extraction tools (can be AI-models) along with the smaller model (e.g. GLM-4.1V-9B-Thinking) acting as comparator. The model then gathers the exercise extractions and if ambiguity exists selects the most common extraction.

7.3 Context Generation Technology

Context generation is not a purely creative task but a constrained transformation: generated narratives must remain aligned with the extracted exercises, available game assets, and pedagogical

intent. This requires both diversity in candidate outputs and strict validation against predefined constraints.

Architectural options:

Architecture	Accuracy	Latency	Extensibility	Decision
Single LLM generation	Medium	Low	Low	✗
Rule-based templates	Low	Very low	Low	✗
Supervisor-worker model	High	Medium	High	✓

Decision:
Single-pass LLM generation lacks reliable accuracy and makes regeneration difficult when constraints are violated. Rule-based systems offer predictability but fail to produce engaging or varied narratives. The supervisor-worker architecture provides a balanced solution by enabling parallel candidate generation while retaining centralized control over correctness and coherence, by assigning agents for both diverse generations and evaluation.

7.4 Game Runtime Technology

The game runtime is responsible for visualizing the effects of student decisions in a way that is both engaging and technically feasible on standard school hardware. The prototype explicitly avoids high-end graphical requirements in favor of clarity, responsiveness, and ease of iteration.

Considered engines:

Engine	Learning curve	Deployment	Ecosystem	Decision
Unity	Medium	Excellent	Very large	✓
Godot	Low	Good	Medium	✗
Unreal engine	High	Heavy	Large	✗

Decision:
While Unreal Engine excels in high-fidelity 3D environments, its complexity and hardware demands make it unsuitable for a lightweight educational prototype. Godot offers an accessible alternative but lacks the same level of ecosystem maturity and institutional adoption. Unity provides the best balance between tooling maturity, cross-platform deployment, and long-term maintainability in an educational context.

7.5 Teacher and IT Coordinator UI

For the teacher and IT coordinator interfaces, a distinction must be made between client-side rendering (CSR) and server-side rendering (SSR).

SSR provides consistent rendering speeds across devices, faster initial page load, more advanced SEO, caching, and reduced round-trip messages at the cost of higher development complexity and server load. CSR provides simpler development, lower server load, faster interactions after initial load, and better support for highly interactive apps at the cost of inconsistent rendering speeds across

devices, slower initial loads for larger projects, weaker SEO, limited caching, and a larger number round-trips.

However, given the scale of the project and its application, neither CSR nor SSR have clear advantages over the other. As the project is relatively small the SSR performance will have only marginal improvements over CSR and SEO is of higher importance for public websites rather than authenticated web applications. This shifts the decision toward framework tooling and ecosystem fit rather than the rendering approach itself.

UI library + Framework Options:

Technology	**Bundle size**[1]	**Performance**	**Ecosystem**	**Familiarity**	**Decision**
React + Next.js	Medium	Medium-High	Large	High	✓
Svelte + SvelteKit	Small	High	Medium	Medium	✗
Vue + Nuxt	Medium	Medium-High	Large	Medium	✗

Decision:
The decision was made to use React+ Next.js due to its extensive ecosystem and high team familiarity. Next.js provides a hybrid SSR/CSR solution with minimal added complexity, providing both faster initial load time and data prefetching, while preserving CSR simplicity and responsiveness.

7.6 Communication Model

The system uses a REST-style API for all communication between the teacher interface, backend services, and the game client. Rather than exposing low-level protocol details, the API is organized around clear domain concepts such as documents, exercises, contexts, quests, and submissions. This keeps interactions predictable, readable, and easy to debug for a multidisciplinary team.

All requests are stateless: each call contains all required information, which simplifies scaling and avoids hidden dependencies. Computationally heavy operations, such as PDF extraction and context generation, are handled as batch processes. This enables progress reporting and ensures bounded, predictable latency for teacher workflows. Teacher actions, student gameplay interactions, and administrative operations are exposed through separate endpoint groups, reducing coupling between different parts of the system.

A typical flow starts with a teacher uploading a PDF, after which the backend returns an identifier for the extraction job. The teacher can retrieve a preview of the extracted exercises and request regeneration if needed. Once contexts are finalized, the game client retrieves quest data. Students submit answers through the game, after which the backend validates the submission and returns an updated world state.

Example:
Extracted exercises are exposed through the /exercises resource. After processing a document, exercises can be retrieved using: GET /exercises?documentId=12345

Each exercise includes the question, correct answer, and a reference to its associated context. Updates made by a teacher are sent back via PUT requests in a similar way.

[1] Bundle size is the total size of all the files (like JavaScript, CSS, images) that are bundled and sent to the browser when a web page loads; smaller bundles usually mean faster load times and better performance for users.

This resource-oriented structure keeps the API consistent with the system's architecture and allows it to evolve alongside the prototype.

7.7 Data Storage and Caching

Educational platforms must manage different types of data, ranging from highly structured and long-lived educational records to short-lived, rapidly changing gameplay state. This raises the question of whether a relational database, a NoSQL database, or a combination of both is most appropriate.

Relational databases provide strong consistency, explicit schemas, and transactional guarantees, which are important for educational data such as exercises, solutions, user progress, and versioned teacher edits. NoSQL databases, on the other hand, are often optimized for flexible schemas and low-latency access, making them attractive for transient or rapidly changing data but less suitable as a system of record.

Based on these considerations, several storage technologies were evaluated with respect to consistency, latency, and suitability for the project's data characteristics.

Technology	**Consistency**	**Latency**	**Use case fit**	**Decision**
PostgreSQL	Strong	Medium	Excellent	✓
MongoDB	Medium	Medium	Lower	✗
Redis only	Weak	Very low	Insufficient	✗

Decision:
PostgreSQL provides strong consistency guarantees for critical educational data, while Redis is used selectively to reduce latency for frequently accessed or transient state. This combination ensures correctness without sacrificing responsiveness.

7.8 Implementation Languages and Runtime Placement

The system is implemented using a small set of well-established programming languages, each selected based on suitability for its specific runtime environment and responsibility within the architecture. This deliberate separation of languages and runtimes supports parallel development, clear ownership boundaries, and long-term maintainability.

The teacher and IT coordinator interfaces are implemented using React with TypeScript and run entirely in the browser. TypeScript provides static typing and improved tooling support, which reduces runtime errors in complex, form-driven workflows such as PDF upload, preview, and correction. Running these interfaces, client-side avoids unnecessary server complexity and aligns well with the usage patterns of internal school tools.

Backend services are implemented in TypeScript or Rust and run on server infrastructure. TypeScript is well suited for rapidly evolving REST APIs and integrates naturally with the front-end stack, while Rust is considered for performance-critical components where memory safety and predictability are paramount. It is also a very interesting upcoming new language that would be worth the try to learn

with this context. This dual-language option allows the team to balance development speed and robustness depending on the specific service.

AI pipelines for exercise extraction and context generation are implemented in Python and deployed on GPU-enabled nodes. Python is the de facto standard for machine learning workflows and offers direct access to modern deep learning frameworks and model tooling. Isolating these pipelines on dedicated GPU nodes ensures that computationally intensive tasks do not interfere with interactive system components.

The game client is implemented in C# using the Unity engine and runs on student devices. This choice aligns with Unity's native language support and allows the game runtime to remain fully decoupled from backend services, communicating exclusively through well-defined APIs and events.

This distribution of languages and runtimes ensures that each subsystem is implemented using technologies that are idiomatic for its domain, while keeping integration boundaries explicit and manageable.

Component	**Language**	**Runtime**
Teacher / IT UI	React + Typescript + Next.js	Browser
Backend APIs	Typescript or Rust	Server
AI Pipelines	Python	GPU Nodes
Game Client	C# with Unity	Student Device

7.9 Relation to Protype Scope

The selected technology stack directly supports the prototype scope defined in Chapter 8 by mapping each core functional requirement to a minimal yet sufficient technological solution.

The exercise extractor is enabled by the hybrid parsing and local LLM pipeline, which allows teachers to upload existing material and receive structured exercises with minimal manual intervention. This directly addresses the need to reduce preparation effort while respecting privacy constraints.

The context generator is realized through the supervisor–worker AI architecture, which enables controlled narrative generation while preserving consistency with extracted exercises and available game assets. This structure supports iteration and regeneration without introducing excessive system complexity.

The game world is implemented using the Unity runtime, which provides the necessary tooling to visualize quests, NPC interactions, and world state changes in a lightweight and accessible manner. This enables students to experience the consequences of their actions without requiring high-end hardware or complex deployment.

Finally, answer validation is handled by backend services that evaluate student submissions and propagate validated results back to the game runtime. This separation ensures correctness and traceability while keeping gameplay logic responsive and deterministic.

Together, these mappings demonstrate that the chosen technologies are not only technically feasible but also tightly aligned with the reduced, proof-of-concept scope of the prototype. Each core feature is

supported by a minimal set of technologies that can be realistically implemented and evaluated within the available timeframe.

The selected technologies directly enable the prototype scope defined in Chapter 8:

[illegible]	[illegible]
[illegible]	[illegible]
[illegible]	[illegible]

7.10 Conclusion

The technology stack presented in this chapter balances privacy, predictability, and feasibility within a realistic academic development context. Rather than optimizing for maximal performance or cutting-edge novelty, the architecture deliberately favors robustness, clarity, and controllability.

By relying on self-hosted, open-weight AI models, the system ensures strict protection of educational content while retaining full control over inference behavior. The use of a lightweight game runtime and event-driven architecture enables meaningful student interaction without imposing excessive hardware or operational requirements. Clear separation between UI, backend services, AI pipelines, and the game client allows development work to proceed in parallel and reduces coupling between subsystems.

Most importantly, the architecture is explicitly designed to support iterative development. Core functionality can be implemented and validated early, while more advanced features such as personalization, extended world persistence, or adaptive difficulty can be introduced later without requiring structural redesign.

As a result, the chosen technology stack is well suited for delivering a functional and evaluable prototype in the next semester, while leaving sufficient room for future evolution beyond the proof-of-concept stage.

8 PROTOTYPE DEVELOPMENT SCOPE AND NEEDS

Next semester will center around actually producing a functional prototype of the solution. Therefore, to ensure a smooth start, it is important to define the irreducible core of our project. This can be realized by prioritizing the most important use cases and mark other use cases as nice-to-haves. Another important aspect is defining the dependencies of the project, finding out what is already available and what needs to be provided before the start of the second semester.

Commented [AL137]: Nice 🤌

Commented [BR137R2]: Hahah yeah 😌

Commented [SS138]: Ik zou dit als een text schrijven, een opsomming is hier overbodig.

8.1 The scope of the project

To ensure that a working product is produced within the allocated time, it is important to determine the core aspects providing the most value to our product and strip them down to their simplest form.
. This approach supports the rapid delivery of a minimal, high-value version and enables early roll-outs, so that feedback can be incorporated into the iterative development process.

Commented [AL139]: @Briek Rombaut alternative

Commented [BR139R2]: Goed verwoord en je mag het gerust zetten in de plaats. Enkel zijn sommige zinnen redelijk lang en zou ik ze opsplitsen

Commented [BR139R3]: Misschien zelfs voor de zin "This not only …. But also …" misschien de opgesomde voordelen oplijsten met bullet points?

Commented [AL139R4]: Kan zeker, doe maar als je wilt, maar maakt het wel langer.

Commented [SS140]: Ik zou dit als een text schrijven, een opsomming is hier overbodig.

The requirement phase in chapter 5 provided the list of use cases and assigned levels of priority to them. The following use cases all had the highest priority assigned:

UC1: As a teacher, I can upload a pdf from which the system extracts exercises pairs.
UC2: As a teacher, I can generate real-life context for my exercises.
UC3: As a teacher, I can edit the generated real-life context.
UC8: As a teacher, I can check the progress of my students individually.
UC9: As a teacher, I can check the progress of my students per class.
UC14: As a student, I can solve quests by talking to NPCs and see the impact of my subjects on the real world through visual effects happening to the corresponding district.
UC17: As the IT-coordinator, I can create an instance for the school.
UC18: As the IT-coordinator, I can create teacher and student accounts for my school.
UC19: As the IT-coordinator, I can configure the class groups at the beginning of the school year.
UC20: As the IT-coordinator, I can assign subjects and teachers to classes.
UC21: As the IT-coordinator, I can edit the class configurations.

However, these highest priority use cases are roughly half of all defined use cases and not all of them are equally important for a working prototype.

Therefore, a more detailed definition of the most important aspects of the project is required to get a clear idea of the core focus of the project.

Commented [AL141]: Formal alternative, shall from here on just mention the informal parts.

8.1.1 Core scope for a proof-of-concept prototype

The core scope will include the 3 most important use cases that were outlined in chapter 5. Another indispensable feature is enabling the teacher to check the process of the individual student as well as the class as a whole. All the interviewed teachers mentioned this as an essential need.

The remaining use cases with the highest priority are all those important to the IT-coordinator. These are less important for a proof-of-concept prototype but would be of big importance if this program would be used in the real world.

Other use cases that were not marked high priority but are important for the correct workflow of the platform are UC13 that lets the student roam around in the game world, and UC16 that enables the student to check their own process.

The core score contains
Hieronder is nog niet aangepast, dus hoef je nog niet na te lezen

To sum up the most important parts of the project:

- Exercise extractor
- Context generator
- Answer checker
- Game world

The feasibility of realizing this core scope is supported by the architectural and technology decisions described in Chapter 7.

The most important aspects of the game world for a minimal working prototype were identified. This version of the game is the simplest version that still captures the essence of the project.
At the most basic level, the game should be able to do the following:

1) Provide building blocks for the context generator, such as NPCs, buildings and several states for the buildings and NPCs. (e.g. NPC is sad/happy. Building is clean/on fire)
2) Convert a structured textual representation of a quest into a playable quest using the building blocks.
3) Allow the student to play a quest. This includes:

- Students can answer questions from NPCs
- The correctness of the answer of a student affects the state of the world (states of the NPCs, buildings decided in the generation of the context).

For a minimal scope, the effects that the player has on the world are only temporary. A quest would be a one-time playable adventure where the effects of the player's decisions are confined to the buildings generated in the quest. Each quest would consist of a path along generated buildings where at each building the player needs to solve a question to pass to the next question. The state of the building or NPC near the player will then change, according to how the student answered the question.

For the student to be able to see the difference in their progress for each course, there is a plan to create a separate world/district per course. But for a minimal scope, a single world/district would suffice. Creating a separate one per course would be more time consuming. Starting with one would allow testing to start sooner.

8.1.2 Nice-to-have enhancements

The ideas discussed in this section are not strictly necessary for the core scope but would enhance our product in some way.
Not all other use cases are mentioned here but are also nice enhancements for the project. As to not list the same things twice, here is a list of personal favourite use cases and features to add.

Commented [SS142]: Dit is geen zin

Commented [SS143]: Zou hier de gradaties in effecten benadrukken.

Commented [SS144]: Dit is nog nergens vermeld als ik het goed heb.

Commented [SS144R2]: Waarmee ik dus bedoel dat je zult moeten uitleggen wat je daarmee wilt zeggen

Commented [SS145]: Opsomming op deze manier op twee niveaus die eigenlijk gelijk staan is verwarrend om te lezen.

Commented [AL146]: The buildings aren't generated, rather the building selection is right?

Commented [AL147]: I don't think this is what we agreed upon as a team. I now understand the choice of words in the previous sentence but this can't be state unless the whole team agreed on a generative approach rather than a predefined environment like a city.

Commented [SS148]: Not sure this is true as covering multiple subjects is what sets us apart from competitors, but I guess we will have to start somewhere either way.

Commented [SS148R2]: So maybe leaving it in is fine

Commented [BR149]: Alternatief voor tekst erboven

Not all ideas will make it in the final version, but as much enhancements as possible will be added in the given time.

8.1.2.1 Enhancements for the game

After implementing the core of the game, the experience of the user could be further enhanced by adding some of the following ideas:

- Customization options for the player character
- A shop and currency to buy cosmetics
- NPCs reward the player with items, which need to be used to advance to next parts of the quest or can be sold for currency
- Adding a hometown for the player that can be decorated as a reward for doing quests.
- The ability to make the stakes higher for a certain quest because it influences the state of the player's hometown

Commented [SS150]: What do you mean? Have we discussed any of these?

8.1.2.2 Enhancements for the school side

The following ideas are not strictly necessary but could make the teacher side more appealing:

- Adding the ability to view user statistics
- Teachers can add their own NPCs and buildings

Commented [SS151]: Do you mean seeing the progress of students? I think this is rather important and not just a nice to have. It was mentioned by interviewed teachers and the publisher.

Making setting up the program as easy as possible is not a priority for a small prototype but would make the product usable in the real world. This is why all the use cases for the IT coordinator would also enhance our product.

Commented [SS152]: Usability of the system is our first system wide quality attribute, so what you state here isn't exactly true.

8.2 The project's needs

An assessment of the needs and dependencies for the project was done, so that the development process will have a smooth start.
Most needs are already covered or can be managed by the team in the second semester. However, a dedicated hosting space on the HPC will need to be provided by the university.

8.2.1 Access to example exercises and solutions

Commented [SS153]: The fact that we're allowed to use the exercises from Plantyn is already mentioned in the Ideate Phase part of the Design Thinking chapter.

One of the requirements is: Access to documents that contain exercises and solutions for STEM (Science, Technology, Engineering and Mathematics) courses in the first and second grade of secondary education.
These exercises are mostly not openly available, but we can get access by:

Commented [SS154]: Would leave this out, by the time they reach this part of the document, I hope they will already know what STEM means haha

- Contacting teachers and asking if we can use their handmade assignments and correction keys.
- Contacting publishers and asking for access to exercises and correction keys used in their student textbooks.

The team already has contact information for a handful of teachers and one textbook publisher.
The publisher has already given approval for using their content and has provided the team with a teacher key to access it.
Some example exercises were also already provided by a secondary math and IT teacher
If access to more exercises would be necessary during the second semester, the team will contact more publishers and teachers.

8.2.2 Open source LLMs availability

To implement the context generator and exercise extractor, there will most likely be built upon existing LLMs.
External LLM APIs cannot be used, because exercises and solutions, which are not our intellectual property, will be used by the models. It should be avoided uploading those to third-party services.
A custom pipeline may need to be built, or model parameters may need to be tweaked.
Because of these reasons, self-hosted LLMs that are open-source or open-weights are the preferred choice.
There exist multiple suitable models that can be run locally and adapted if necessary.
The concrete selection and evaluation of these models is discussed in Chapter 7.

Commented [SS155]: I feel like a lot of this should go in the technology chapter and not here

Commented [BR155R2]: Yeah that might be more fitting.

@Francisco Van Langenhove what do you think? This concerns the technology chapter that you worked on

Commented [FL156]: @Briek Rombaut, zo chaud?

8.2.3 Reserved hosting space with GPU access

A hosting space will be needed with GPU access. This will be used for self-hosting our LLMs and running the server-side of our program. This hosting space would need to be available, continuously throughout the duration of the second semester.

The university's HPC could be used for this. This would mean that the university would need to reserve resources of the HPC for the duration of the whole second semester.

Students can request access themselves to the HPC infrastructure, but this is limited to maximum reserved periods of 72 hours.
This is why the professors for this course will probably need to provide a hosting space dedicated to the project for the course of the second semester.

The availability of persistent GPU resources is a prerequisite for running local AI models and backend services.
How these resources are utilized within the system architecture is further detailed in Chapter 7.

Commented [PV157]: @Briek Rombaut

Commented [BR157R2]: Bedoel je dat je dit zou toevoegen aan het bestaande deel op de plaats waar je het hebt geplaatst? Dan ist goed

---

8.2.4 Game engine license

Since none of the team members have prior professional experience in game development, it is important to rely on an engine that is well-documented, widely used, and supported by a large developer community. This lowers the learning threshold and reduces implementation risk during the prototype phase.

The game engine *Unity* seems the best fit, because it is a mature engine with lots of documentation and is widely popular among game developers. These facts make it significantly easier to find necessary information or partially re-usable code examples.

Unity is not free for all purposes and has a lot of different subscription plans. One of those is the *Unity Student plan*, which seems the most applicable for our project. This is a free plan for verified students.

Commented [FD158]: Zou dit eerder verwoorden als: "We plan to use the game engine Unity, because it has extensive documentation and is popular among game developers. These facets make it significantly easier to find necessary information or partially re-usable code examples online."

Commented [BR158R2]: Zo goed?

Commented [FD158R3]: "for this" vind ik nogal vaag en overbodig

Since the University of Gent is already recognized as an educational institution by *Unity*, it is possible for students to request a *Unity Student* licence themselves using their UGent email.

9 USE OF AI

Commented [FD159]: this always needs to be at "very end of deliverable", right before appendix and references

Generative AI was used in a limited and carefully controlled way throughout the project. The tools employed were primarily ChatGPT, with Copilot used by some members during the creation of visual variations for the first prototype. Most team members used AI only sparingly, mainly to adjust wording in the deliverable or to generate alternative background images. All AI-generated output was critically reviewed before inclusion, and many tasks were intentionally performed manually to ensure originality and reliability.

A smaller number of team members made broader use of AI as a supportive tool. For these members AI functioned as a creative partner that helped structure ideas, summarise research findings, compare the concept with existing alternatives, and explore variations of the problem statement. AI was particularly useful for handling large amounts of context and highlighting connections that are less immediately visible to humans. Across the group there was consistent awareness that AI is not an authoritative source, and all generated material was verified against reliable references and coach feedback. As a result, AI contributed to efficiency and clarity without compromising the integrity of the work.

REFERENCES

Commented [SS160]: TODO before handing in fix this table!!